# Facing Change Again: Meta-Analyses of Gender and Climate Change Attitudes Worldwide

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With the increasingly pressing need to address climate change comes a similar need to understand the factors that undermine public support for mitigation. One prominent recent analysis of three cross-national surveys argues that people—especially men—in richer countries express less concern for climate change and see more costs and fewer benefits to mitigation, pointing to compensation as a potential solution. This reassessment employs original meta-analyses of hundreds of surveys to reveal that while gender differences in climate concern are indeed larger in richer countries, both women and men express more concern in such settings relative to poorer countries, and there is no relationship between economic development and gender differences in mitigation's perceived costs and benefits. Additional meta-analyses indicate support for climate change policy is higher in richer countries among both women and men, further substantiating that patterns of climate attitudes are very different than previously understood.

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Climate change and its devastating consequences for humanity have been well understood among scientists for many decades, yet no country has adopted policies commensurate with the scale of the threat. Because policymaking to fight climate change depends on public opinion that recognizes the problem (Bromley-Trujillo and Poe 2020), identifying the sources of attitudes toward climate change is crucial. One prominent recent article, Bush and Clayton (2023), builds on findings that women in the United States and Europe express more concern about climate change than men (see, e.g., Egan and Mullin 2017, 215; Ergun, Karadeniz, and Rivas 2024, 9) to advance our understanding of this increasingly pressing issue. It makes three claims, each building on the last. The first claim is that the gender gap in climate change concern is larger in richer countries than in poorer ones. The second is that this wider gap is due to lower levels of concern in richer countries than in poorer countries, levels that are especially low among men. The third claim is that these patterns arise because people in richer countries, and especially men, perceive fewer benefits and more costs of policies designed to mitigate climate change. Drawing on three cross-sectional surveys and a paired case study of focus groups in Peru and the United States, Bush and Clayton (2023) finds evidence to support each of these claims. Though untested directly, the three claims are bound together and motivated by the argument that support for climate mitigation policies similar declines with economic development, and especially so among men.

This reassessment assembles a comprehensive collection of data on climate change attitudes comprising over 3000 national surveys and reexamines these three claims and the argument by drawing on meta-analysis tools growing in popularity in political science. This much larger evidentiary base confirms that the difference in climate change concern between men and women is generally larger in richer countries. But relatively few surveys show men in richer countries expressing lower levels of climate change concern than men in poorer countries—in fact, consistent with postmaterialist theory (see, e.g., Inglehart 1977, 2018; Inglehart and Welzel 2005), meta-analysis of the available cross-national surveys indicates the opposite is the case. Meta-analyses of surveys directly asking respondents to assess the costs, benefits, or both of policies fighting climate change show little to no differences between men and women in these attitudes at any level of economic development. Moreover, meta-analyses of the available survey data show that support for policies fighting climate change is *higher* in richer countries and that gender gaps in sup-

port are only slightly larger in more economically developed countries. Together, these results suggest that gender and economic development play very different roles in the formation of climate change attitudes than those indicated in Bush and Clayton (2023). More generally, they caution against relying overmuch on the evidence provided by just one or a few surveys.

# 1 Gender Differences in Climate Change Concern

The first of the three claims made in Bush and Clayton (2023, 592) has to do with "variation in the climate gender gap." Specifically, it holds that the difference between women and men in climate change concern is larger in richer countries than in poorer ones. The main models of Bush and Clayton (2023, 593, Table 1, Models 1 and 4, see also 600, Figure 4) estimate the following equation for the climate concern of individual $i$ in country $j$:

$$\begin{aligned} Concern_{ij} =& \gamma_{00} + \gamma_{01} Log(GDP/capita)_j + \gamma_{10} Women_{ij} \\ &+ \gamma_{11} Log(GDP/capita)_j \times Women_{ij} + r_{0j} + r_{1j} Women_{ij} \end{aligned} \tag{1}$$

That is, they are two-level hierarchical models with varying intercepts for each country ($r_{0j}$), varying slopes for the coefficient for the gender of the respondent in each country ($r_{1j}$), and a cross-level multiplicative interaction between country economic development and respondent gender (captured by $\gamma_{11}$). Positive values of $\gamma_{11}$ indicate support for this first claim. And the estimates of $\gamma_{11}$ are in fact positive in models of one item on climate concern from the 2016/2017 AmericasBarometer (Model 1), and one item from the 2015 Pew Global Attitudes Survey (Model 4), as well as one item from a ten-country survey conducted by the work's authors in 2019-2020 (600, Figure 4), providing evidence that the difference in climate concern expressed by women and men is larger in richer countries.

The three surveys analysed in Bush and Clayton (2023), however, represent only a tiny fraction of the available surveys that asked respondents questions on how concerned they are about climate change. Given the importance of the topic, it should come as no surprise that questions on climate concern have been included in many cross-national surveys, including not only four waves of the AmericasBarometer and seven Pew Global Attitudes

Surveys, but also such well-known surveys as the Arab Barometer, the Eurobarometer, and the World Values Survey. In fact, 107 different surveys of at least ten countries asked 89 different items on concern for climate change, a total of 265 survey-items altogether (details on these survey-items and the procedure employed to identify them are available in the online supplementary materials, Section A).

The large number and wide variety of these survey items makes their collective analysis difficult. This is a common issue. Despite long-standing advice to maximize the evidence employed to test theories (see, e.g., King, Keohane, and Verba 1994), survey research frequently takes advantage of the data available in only one or just a few surveys. This is understandable given the difficulty of incorporating the evidence provided by different items from different surveys. Simply presenting and interpreting separate analyses of each of the available survey-items is straighforward, at least superficially, but the feasibility of this approach collapses as the number of survey-items increases and the results their analysis provide diverge. Pooling multiple surveys provides a means to bring more data to bear, but it depends on identical items and response categories across the surveys: a long literature on question wording effects cautions against pooling even such similar questions as "How concerned are you about climate change?" (asked in the Life in Transition Survey, Wave 2) and "How concerned are you, if at all, that global climate change will harm you personally at some point in your lifetime?" (asked in the 2015 Pew Global Attitudes Survey) together in a single analysis (see, e.g., Rugg 1941; Krosnick, Malhotra, and Mittal 2014; Solt et al. 2016). A diversity of items enables a variety of techniques to minimize measurement error—including principal components analyses (PCA), item response theory models (IRT), and weighted scales indices (WSI)—but these methods of combining multiple items into a single dependent variable depend on those items being answered by the same respondents (see, e.g., Stoetzer, Zhou, and Steenbergen 2025; Fu and Green 2026). Recent contributions to the study of comparative public opinion have dealt with the issue of incomparability across surveys and items by estimating latent variables (see, e.g., Claassen 2020; Woo, Goldberg, and Solt 2023; Woo et al. 2025); these methods, however, typically generate measures at the country-year level and so prevent examination of individual-level variables like gender (but see Caughey, O'Grady, and Warshaw 2019).

Meta-analysis supplies a different approach to resolving these complications. Meta-

analysis is a tool to "consolidate the overall effect size" from many analyses (Godefroidt 2023, 25). Meta-analysis has not been frequently employed in political science until recently—Arel-Bundock et al. (2026, 38) counts just 85 articles employing the method across 141 journals in political science and closely adjacent fields through 2021—but it is growing increasingly common in the discipline (see, e.g., Kołczyńska 2025; Weber et al. 2025; Barceló forthcoming). With good reason: incorporating the evidence provided by many studies, meta-analysis provides a means to resolve discrepancies and come to conclusions grounded in all of the available evidence. Moreover, meta-analysis yields the same results as a properly-specified pooled model of the original individual-level data (when assessing an existing literature also known as a 'meta re-analysis' or a 'mega-analysis') while being more straightforward to implement and interpret (see, e.g., Burke, Ensor, and Riley 2017, 855; Stahlmann et al. 2026, 548–49).[1]

Although typically the analyses to be consolidated have already been published in many different articles (see, e.g., Schmid, Stijnen, and White 2020), one can also conduct meta-analysis to cumulate the evidence provided by one's own original analyses within the same manuscript, a procedure that Goh, Hall, and Rosenthal (2016) labels a "mini meta-analysis" and McShane and Böckenholt (2017) calls a "single-paper meta-analysis (SPM)." Political scientists have recently begun to take advantage of this latter technique. Pérez, Vicuña, and Ramos (2024) presents a mini meta-analysis of five original experiments on solidarity among people of color that vary the ethnic group of the sample and treatment, and Gazmararian and Milner (forthcoming) uses meta-analysis to summarize the results of four separate multilevel models of different climate shocks' effects on the likelihood that respondents to the 2019 World Risk Poll considered climate change to be among the top two risks to their safety in their daily lives. Pérez, Lee, and Luna (2025) analyses data from each of seven surveys on the relationship between individuals' prioritization of racial or American identification and their partisanship that operationalized the former concept differently, presents the results of each of these separate analyses, and offers a mini meta-analysis that combines those results. This last research design points toward the solution offered here: when many surveys include different items operationalizing the

[1]These works note that meta-reanalysis does have important advantages over standard meta-analytic techniques, including the opportunity to respecify and standardize published analyses, to maximize statistical power when sample sizes are small, and to mitigate selective reporting and publication bias (see also, e.g., Graham forthcoming; Sarkar and Coppock forthcoming). Meta-analyses of original analyses, however, share these advantages.

same concepts, perform original analyses of all of the available surveys, and use meta-analysis to synthesize the results. As the novel analyses to be collectively assessed in the present work number into the hundreds and even thousands, 'single-paper' is a better descriptor than 'mini' for the meta-analytic technique employed here. Such large-scale SPMs can be seen as belonging to a family of approaches to aggregating many analyses that also includes Bayesian model averaging (BMA) (e.g., Montgomery and Nyhan 2010) and multiverse analysis (e.g., Saraceno, Hansen, and Treul 2021), but whereas the full space over which BMA consolidates results is defined by the inclusion or exclusion of each of many possible control variables and that of multiverse analysis a wide range of alternative measures and modeling choices, large SPMs synthesize results over different items in many surveys.

The SPM employed to assess the evidence for the first claim of Bush and Clayton (2023) proceeds as follows. First, mirroring exactly the main analyses of Bush and Clayton (2023), Equation 1 is estimated for each of the 265 cross-national survey-items of climate concern, with logistic regression employed for responses to dichotomous survey items, ordinal logistic regression for survey items with three or four ordered response categories, and linear regression for items with five or more ordered response categories. In the language of meta-analysis, each of these analyses is a study. Next, following standard meta-analytic practice (see, e.g., Schmid, Stijnen, and White 2020, 35; Godefroidt 2023, 25), the estimate of the coefficient for the change in the difference across genders with economic development (that is, $\gamma_{11}$) from each study is converted to a correlation to impose a common scale, and because the sampling distribution of correlations is skewed, the Fisher's $z$ transformation is applied to each correlation to yield a normal distribution of effect sizes. Finally, these effect sizes are themselves modeled in a Bayesian multilevel model that includes varying intercepts for each survey (as the results for different items in the same survey are not independent) and each distinct survey item (as the results for the same item asked in different surveys are not independent); the use of a Bayesian approach allows the uncertainty in the correlations due to sampling error to be seamlessly incorporated into the model.

These results are plotted in Figure 1. The left panel displays the distribution of $t$-values for the coefficients of the interaction between gender and logged GDP/capita across studies. The findings of these studies are not consistent. Some studies yield negative values,

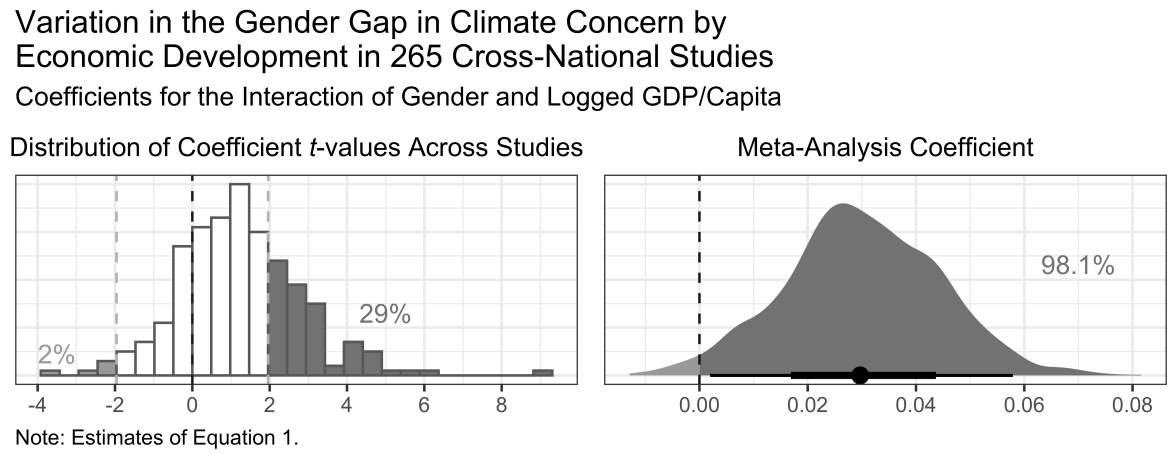



Figure 1: The Difference Between Women and Men in Climate Concern Is Generally Larger in Richer Countries Than in Poorer Countries

indicating a *smaller* climate gender gap in richer countries; in 2% of the studies, this negative result even reaches statistical significance at the 95% level. Most studies do give positive values for $\gamma_{11}$, though only in 29% of the studies do the coefficient estimates reach statistically significance at that level. Were a researcher to examine only one or a few survey-items, as in Bush and Clayton (2023), the conclusion reached would depend on which surveys and items were analyzed. Even with the full distribution of study results, it is not immediately evident whether the claim that the climate gender gap is larger in richer countries than in poorer ones is supported by the available evidence: about seven in ten studies yield a null result.

The right panel of Figure 1, however, displays the posterior probability distribution of the meta-analysis coefficient that consolidates all of these findings. It indicates that, taking into account the dependence between studies of items in the same survey and studies of the same item in different surveys (and, of course, conditional on this collection of the available data as well as the uninformative, uniform prior), the probability that the correlation between whether a respondent is a woman and the climate concern that respondent expresses is larger at higher levels of economic development is over 98%. This result provides support for the finding of the main models of Bush and Clayton (2023) for that article's first claim: women express increasingly more concern about climate change than men where GDP/capita is higher. As this conclusion is based on a comprehensive set of cross-national survey-items on the topic, an evidentiary base roughly 90 times the size of that presented in Bush and Clayton (2023), we can now be much more confident

that this first claim is supported by the available evidence.

## 2 Economic Development and Climate Change Concern

The second claim of Bush and Clayton (2023) is that the wider differences in concern among men and women in richer countries arise because "*both* men and women will be less concerned about climate change in wealthier countries than in poorer countries, but that men's concern will decline more rapidly" (Bush and Clayton 2023, 591). The article's analyses of the three cross-national surveys it examined each indicate climate concern is lower in richer countries and particularly among the men of those countries (Bush and Clayton 2023, 593–94 and 600).

This claim and these findings are both surprising. One of the most influential theories in comparative public opinion over the past half century, Inglehart's postmaterialism thesis, holds that as countries grow richer, the values of their citizens will shift beyond worries for order and immediate material security to broader concerns for individual autonomy and quality of life (see, e.g., Inglehart 1977, 2018; Inglehart and Welzel 2005), notably including the protection of the environment (see, e.g., Inglehart 1995; Welzel 2013). That is, the postmaterialism thesis would lead us to expect the opposite of the claim put forward in Bush and Clayton (2023), that climate concern should be expected to be higher in richer countries than poorer ones.

Which theory is supported by the available evidence? The interaction term in Equation 1 means the estimated conditional effect of economic development on climate concern depends on the gender of the respondent as follows:

$$\frac{\partial Concern_{ij}}{\partial Log(GDP/capita)_j} = \gamma_{01} + \gamma_{11} Women_{ij} \tag{2}$$

Estimates of this conditional effect are calculated for women and for men from each of the 265 studies described above and similarly converted to correlations with the Fisher's $z$ transformation applied. And again, these correlations are themselves modeled in a Bayesian multilevel model with varying intercepts for each survey and for each survey item that also incorporates the uncertainty in the correlations due to sampling error.

Figure 2 shows the findings. The panels on the left side display the distribution of $t$-values of the coefficients for logged GDP/capita in these analyses, with the top and

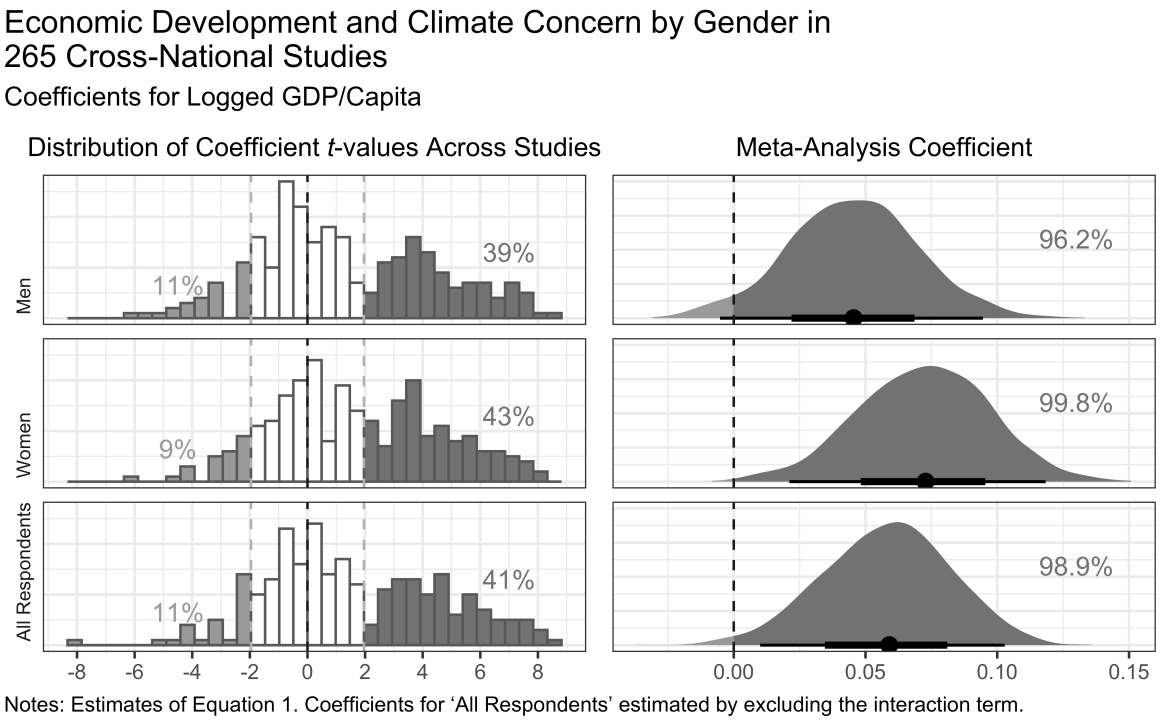



Figure 2: Both Men and Women Tend to Express More Concern About Climate Change in Richer Countries Than in Poorer Countries

middle panels showing estimated conditional effects for men and for women calculated using Equation 2 and the bottom panel the estimated effect on all respondents when the interaction term is omitted from the model. Only about one in ten of these cross-national analyses evince a statistically significant negative relationship between economic development and concern for climate change similar to those presented in Bush and Clayton (2023). On the contrary, most estimated effects are positive, and a large share are both positive and statistically significant at the 95% level.

As in the previous section, meta-analysis provides the means of reaching conclusions based on all of the available evidence. The resulting posterior probability distributions of the meta-analysis coefficient for logged GDP/capita are shown in the right-hand panels of Figure 2. The vast bulk of each is positive: in even the smallest, the estimated effect of economic development for men, 96.2% of the posterior is positive. Consistent with the postmaterialism thesis—and contrary to the second of the claims presented in Bush and Clayton (2023)—both women and men in richer countries generally express *more* concern about climate change, not less, than people in poorer countries.

# 3 Gender, Economic Development, and Perceptions of the Benefits and Costs of Mitigation

The third claim Bush and Clayton (2023, 597–98) advances is that in richer countries relative to poorer countries, both men and women, but particularly men, perceive themselves as having less to gain and more to lose from the major economic and social changes required to mitigate climate change. It is their aversion to these losses, that article asserts, that leads them to express less concern for the problem. The evidence that work presents is based on text analyses of open-ended questions in an original ten-country survey and of focus groups conducted in the United States and Peru; it puts forward no evidence based on more typical survey items. Putting aside for the moment that the meta-analyses above show that people in richer countries do *not* express less concern for climate change, we turn then to an examination of the available evidence on how the perceived benefits and costs of decarbonization vary with economic development and gender.

For the present work, 20 cross-national surveys were identified that include one or more items that tap perceptions of the costs, benefits, or both of climate change mitigation that were asked in ten or more countries (details are presented in the supplementary materials, Section A). For example, the 2022 Risks That Matter survey (OECD 2023) asked how much concern respondents felt regarding "rising energy/fuel costs" due to "building up green infrastructure, emissions limits and carbon taxes," while Eurobarometer 95.1 asked respondents their agreement with the statement, "The costs of the damages due to climate change are much higher than the costs of the investments needed for a green transition" (European Commission and European Parliament 2023).

As in the foregoing examination of climate concern, multiple items in these cross-national surveys are each analysed as a separate study in a multilevel model parallel to the model of climate concern described by equation Equation 1. This yields 63 studies in all. Following the same procedure described above, the Fisher's-$z$-transformed correlation between the dichotomous gender variable and perceptions of the benefits versus costs of mitigation is calculated for each of these studies. The responses to all of these items were recoded so that higher values indicate perceptions of greater benefits and/or lower costs of climate change mitigation (see the supplementary materials, Section A, Table SM2). Therefore, if perceptions of the costs of climate change mitigation relative to its benefits

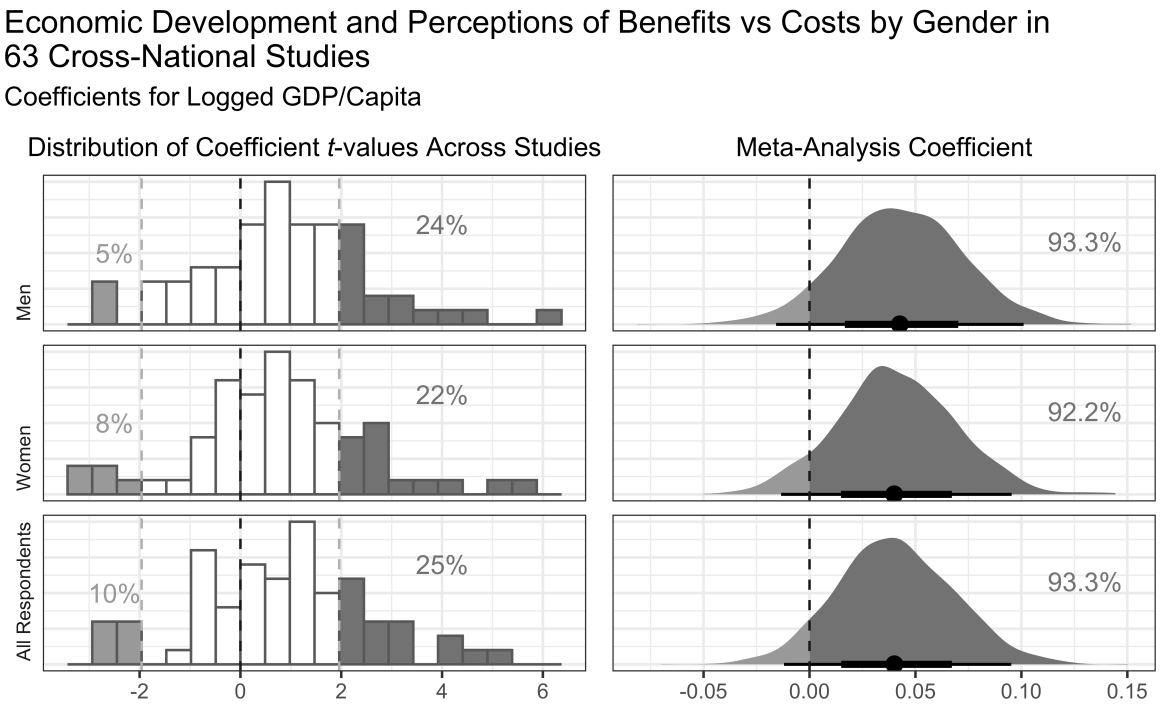



Figure 3: Perceptions of Mitigation's Benefits vs Costs Generally *Increase* with Level of Development

are greater where economic development is higher, as Bush and Clayton (2023) argues, the coefficients for logged GDP/capita by gender will be negative. And if this pattern is especially pronounced among men, the coefficient for the interaction between the indicator variable for a woman respondent and logged GDP/capita will be positive.

Figure 3 displays the results for the conditional effects of logged GDP/capita. They contradict the theory proposed by Bush and Clayton (2023). Neither men nor women in richer countries consistently express less perceived benefits or higher costs than their counterparts in poorer countries; to the contrary, in both cases over 90% of the posterior probability distribution is positive. Given the greater concern for climate change among those living in richer countries seen in Figure 2, it is perhaps not surprising that those in richer countries hold a more favorable perception of the tradeoffs involved in mitigating the problem as well.

That the relationships between the context of economic development and both women's and men's perceptions of these tradeoffs are both positive does not imply that any gaps between women's and men's perceptions are constant over economic development's range. Still, the evidence from the cross-national studies generated with this collection of survey-items suggests that this is the case. The top left panel of Figure 4 shows that only in

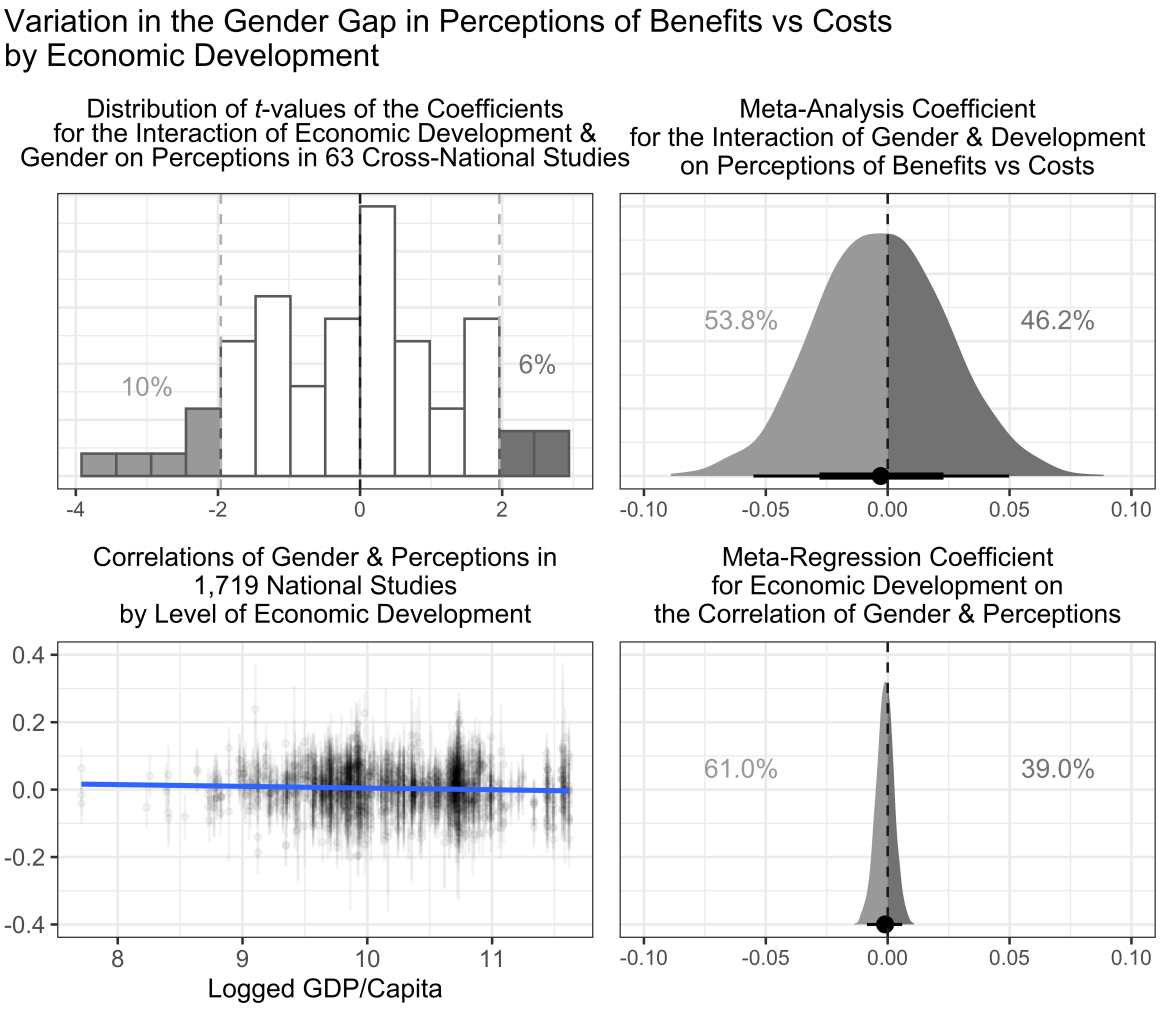



Figure 4: The Available Survey Evidence Does Not Indicate That Gender Gaps in Perceptions of Mitigation's Benefits vs Costs Increase with Economic Development

four of the 63 studies provide support for Bush and Clayton's (2023) claim that the gender gap in perceptions of climate mitigation's costs and benefits is wider at higher levels of economic development, while six studies actually indicate statistically significant declines in the gender gap. The top right panel displays the results of a meta-analysis of these cross-national studies, again with varying intercepts for each survey and item: less than half of the posterior probability distribution of the meta-analysis coefficient of the interaction of gender and logged GDP/capita is positive, and its median is very close to zero.

The relatively small number of cross-national studies available for this SPM, however, may raise concerns about the statistical power of the meta-analysis to distinguish a true effect from zero. One way to increase the number of observations and hence sensitivity is by changing the unit of analysis used to investigate the question (King, Keohane, and Verba 1994, 219–23). Here that can be done by treating each country-year-survey-item as a study of the relationship between gender and climate attitudes and then using meta-regression—which models, rather than summarizes, differences across studies (see, e.g., Higgins, López-López, and Aloe 2020)—to test whether the differences across these studies can be explained with differences in economic development.

The results of this approach are presented in the bottom panels of Figure 4. The bottom left panel is a scatterplot with the (transformed) correlation between the indicator for women with perceptions of the benefits versus costs of climate mitigation in the country-year-survey-item studies on the $y$-axis and GDP/capita for the country-years on the $x$-axis; dots indicate point estimates and the vertical whiskers the uncertainty due to sampling error. The relationship appears to be flat, perhaps slightly negative.

The bottom right panel presents the posterior probability distribution of the coefficient for logged GDP/capita on these correlations from a four-level meta-regression with varying intercepts for each country, survey, and item. As expected, this distribution is much more tightly bounded than that of the analysis presented in the top panels, in which each study is a cross-national survey, giving a much more precise estimate of the relationship between economic develpment and attitudinal gender gaps. Nevertheless, it yields a very similar conclusion: its median value is very close to zero, providing no support for the argument that gender differences in these perceptions are shaped by levels of economic development. There is no evidence for either part of Bush and Clayton's (2023) third

claim.

# 4 Gender, Economic Development, and Support for Climate Policy

Bush and Clayton (2023, 591) motivates its three claims with the argument that gendered differences in perceived "costs and benefits shape whether individuals *support decarbonization* and are concerned about climate change" (emphasis added). Although that article's analyses do not directly address it, support for decarbonization is arguably this argument's most important and policy-relevant component. Advances in policymaking on contentious issues like climate change depend on permissive public opinion (see, e.g., Lax 2009); moreover, efforts toward decarbonization are both highly visible in and tangible to people's lives, just the characteristics that make policy feedback in public opinion—that is, potentially, backlash—most likely (see, e.g., Pacheco 2013). Indeed, it is for these reasons that the work suggests one implication of its findings is that richer countries should compensate their male citizens for the loss of such activities as "driving low-mileage vehicles or eating carbon-intensive foods" to "generate support for decarbonization" (Bush and Clayton 2023, 597, 606). The very different patterns of concern for climate change and perceptions of its costs and benefits than those theorized by Bush and Clayton (2023) revealed in the meta-analyses presented above raise the question of how gender and economic development shape support for policies fighting climate change.

This question, too, is amenable to study with meta-analytic tools and the available cross-national survey data. A total of 77 items asking respondents their level of support for policies fighting climate change fielded in ten or more countries in 20 cross-national surveys were identified (see the supplementary materials, Section A). Again, each of these cross-national survey-items is initially treated as a study, and the results of all of these studies are consolidated using meta-analysis as previously.

We first examine Bush and Clayton's (2023) expectation that support for climate policy is lower in richer countries with more to lose in the economic transformations required by a green transition. Figure 5 shows the results for men and women separately and together. Although the left panels show that some cross-national surveys do show the patterns that Bush and Clayton (2023) predicts, they are relatively few and outnumbered

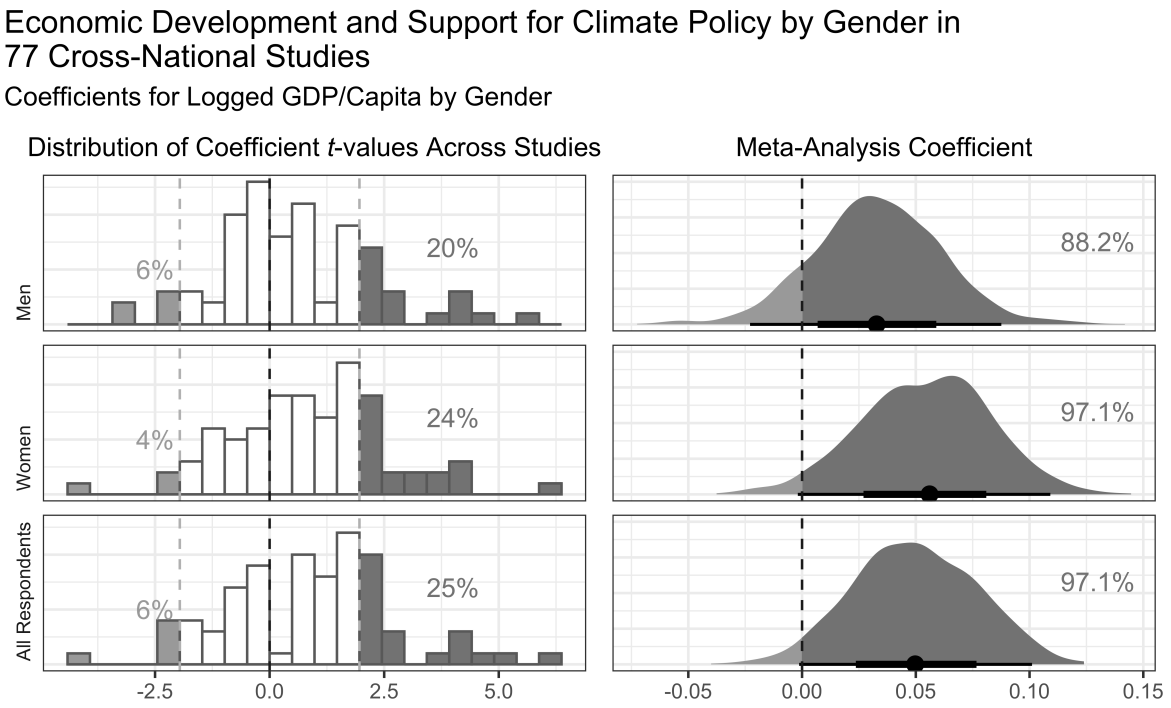



Figure 5: Support for Climate Mitigation Policy Generally *Increases* with Level of Development

by the studies that show instead statistically significantly higher support for climate policy where economic development is higher. The right panels show the posterior distributions of the meta-analysis coefficients that consolidate the results of all of these studies. They indicate that support for climate policy is very likely higher, not lower, in richer countries than in poorer countries among both women and men.

Figure 6 turns to how gender gaps in climate support vary by economic development. In the top right panel, it shows that the median value of posterior of the meta-analysis coefficient for the interaction between gender and develpment in the cross-national studies is positive, but that nearly a fifth of the distribution lies below zero. As in the examination of perceived costs and benefits, greater statistical power is gained by instead using national survey-items as studies and performing a meta-regression predicting their differences with GDP/capita. The lower right panel shows that the posterior distribution indicates that this relationship is definitively positive in these data, but very small, only roughly a third the size of the corresponding coefficient for climate concern (compare this figure with Figure 1 as well as Figure SM2 in the supplementary materials).

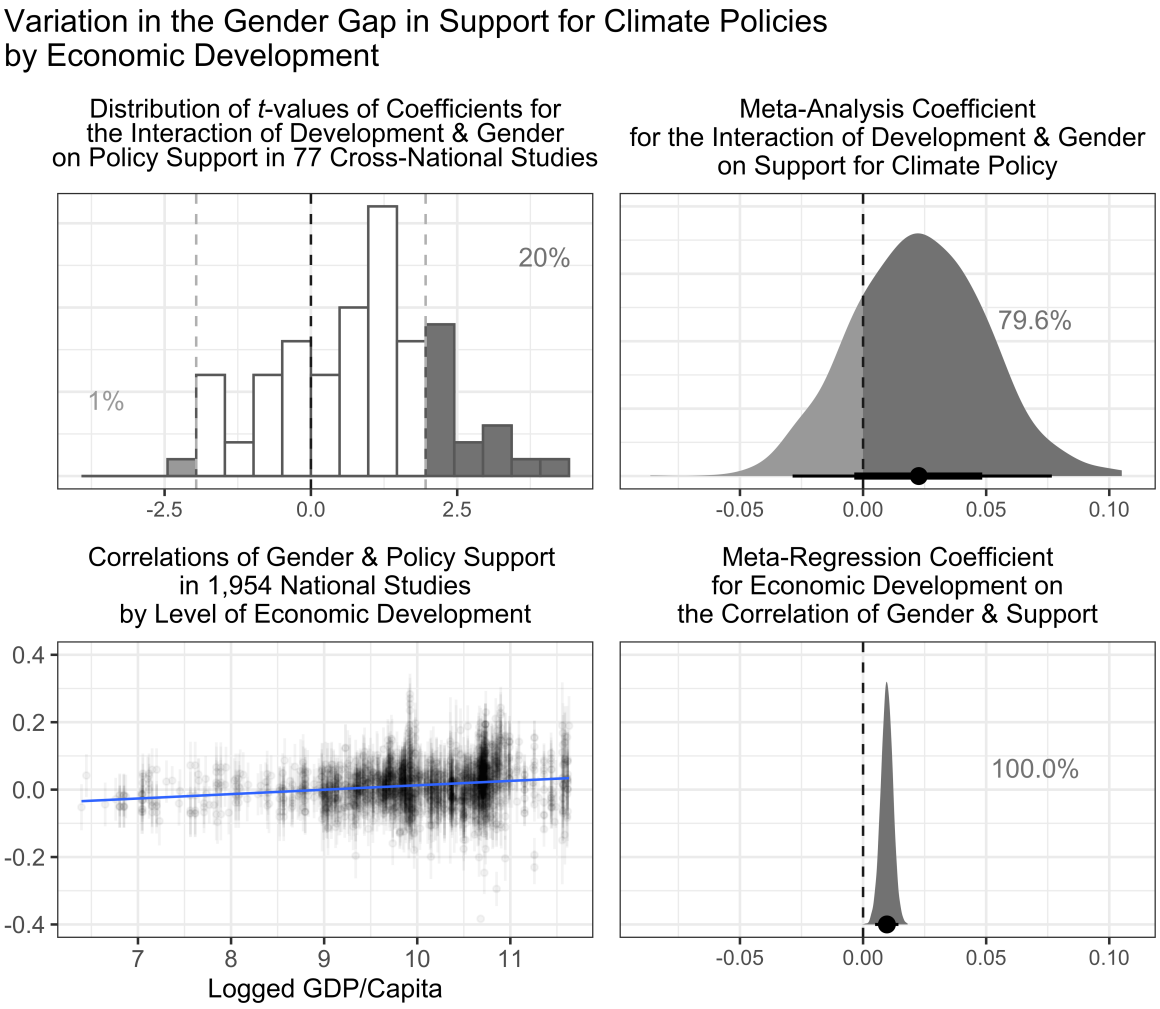



Figure 6: The Difference Between Women and Men in Support for Climate Policies Is Generally *Slightly* Larger in Richer Countries Than in Poorer Countries

# 5 Conclusions

Heeding the long-standing admonition to maximize the evidence employed to test our theories (see, e.g., King, Keohane, and Verba 1994), this reassessment tested the three claims on gender and climate change attitudes and the argument underpinning them that were put forward by a prominent recent work, Bush and Clayton (2023), with large-scale single-paper meta-analyses of a collection of cross-national surveys many times bigger than the three examined in that article. The first claim, that differences between men and women in climate change concern are larger in richer countries, is supported across the hundreds of studies presented here. But, contrary to the second claim and consistent with postmaterialist theory, this additional evidence indicates that levels of climate change concern are *higher*, not lower, for both men and women in richer countries. The third claim, that in richer countries men increasingly see more costs and fewer benefits than women in climate change mitigation, finds no support in the entirety of the available evidence; in fact, perceptions of the benefits versus costs of mitigation rise with economic development. Tests of the argument that support for climate policy should be lower in richer countries and especially among the men living there yielded results indicating that, like concern and perceptions of benefits and costs, support is instead higher—though gender gaps are very slightly larger—in these contexts.

These findings have several important implications. One is that further research is needed to better understand the sources of climate change attitudes—concern about climate change, perceptions of the benefits and costs of addressing it, and support for climate policy—as well as their interrelationships. For one example, the origins of the pattern of larger differences in climate change concern by gender in richer countries proposed by Bush and Clayton (2023) and reinforced by the evidence presented here remain unclear. This same pattern exists in a more muted form in support for climate policies, but as gender gaps in perceptions of costs and benefits if they exist are *not* consistently larger in more economically developed countries, the Bush and Clayton (2023) argument that it is these perceptions that drive this pattern is not supported in the available evidence, and the article's theory that support for fighting climate change is lowest among those with the most to lose is not supported at all. Explaining the interrelationships among these attitudes in more detail is beyond the scope of this reassessment, but that over two dozen cross-national surveys include items tapping at least two of these three concepts

provides the basis for one possible route forward.

Another implication concerns the policy prescriptions put forward by Bush and Clayton (2023). As mentioned in the previous section, that article recommends that richer countries generate support for fighting climate change by compensating men to abandon habits like driving oversize trucks and eating beef (Bush and Clayton 2023, 597, 606). As the available evidence does not support the argument underpinning this recommendation, such a strategy is unlikely to help; in fact, by highlighting these 'losses,' compensation schemes directed at men may even increase men's opposition (cf. Walker, Butland, and Connell 2000). It would seem that those working to devise compensatory policies to mollify concentrated opposition to decarbonization would be better advised to focus potential compensation on opponents who are widely regarded as deserving, like coal workers (see Arel-Bundock and Pelc 2026). In any case, and in light of the urgency with which climate change must now be addressed, efforts toward building public support for decarbonization should be based on solid empirical foundations.

A final implication of this work is methodological, and it has broad applicability to studies of public opinion. It is that meta-analysis is more than simply a method for reviewing a well-developed literature; it is also a means of maximizing the evidence brought to bear on even novel questions of interest. The growing availability of both cross-national survey datasets (see, e.g., Smith 2015) and tools to programmatically wrangle them (see, e.g., Solt, Hu, and Tai 2019) mean that data constraints alone rarely justify drawing on only one or a few surveys. After a more comprehensive body of relevant data is identified, large-scale single-paper meta-analysis provides a straightforward method for incorporating it into a single coherent investigation. Our understanding of political phenomena will benefit greatly if we make more use of it.

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# Facing Change Again: Meta-Analyses of Gender and Climate Change Attitudes Worldwide

Supplementary Materials

## A Data Collection and Coding

This appendix describes the procedure used to select the survey items employed in this article in accordance with PRISMA ('Preferred Reporting Items for Systematic Reviews and Meta-Analyses') guidelines. Though these guidelines were designed for the review and meta-analysis of existing literatures, they are useful for describing the procedure employed to identify and collect relevant survey items in the large-scale single-paper meta-analyses (SPMs) of data on novel questions such as those reported in this article.

### A.1 Search Terms and Databases

Items on climate change attitudes asked in cross-national surveys were identified by a series of searches for the terms "climate", "global warming", "greenhouse", and "carbon dioxide." First, a search of the codebooks of the surveys collected in the `DCPOtools` R package (Solt, Hu, and Tai 2019) was performed. This yielded 406 survey-items. Subsequent searches of the data archives at GESIS, the Pew Research Center, the ICPSR, and the UK Data Service yielded four additional surveys—three recent Eurobarometer surveys and an early survey of the Pew Global Attitudes Project—that had yet to be included in `DCPOtools`, with a total of fifteen survey-items. Finally, web searches for these terms (each with "survey" added) identified the series of nine European Investment Bank Climate Surveys that include a total of 62 survey-items.

### A.2 Inclusion and Exclusion Criteria

The inclusion and exclusion criteria for this study were as follows. Relevant climate survey-items are those pertaining to climate concern, the perception of benefits and costs of mitigation, and support for policies fighting climate change. Survey-items relevant to climate concern were identified as those asking respondents about their level of concern or worry (e.g., "how concerned are you about climate change"); their assessment of the seriousness or threat (e.g., "how serious a problem do you think climate change is at this moment"); as well as items asking respondents to identify the most serious or important problems or challenges for their country, an international organization such as the European Union or United Nations, or the world (e.g., "in your opinion, which of the following

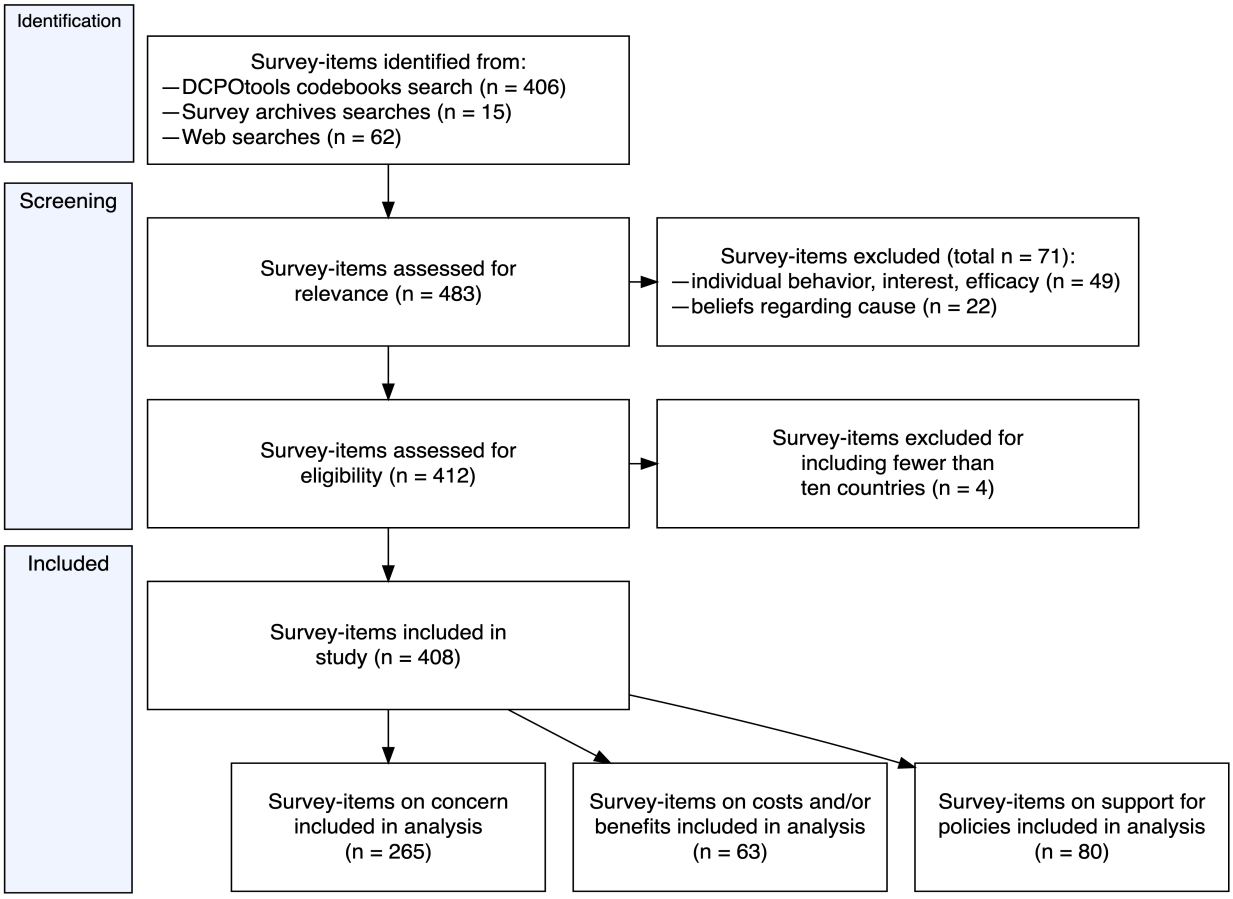



Figure SM1: PRISMA Survey Item Selection Process

do you consider to be the most serious problem currently facing the world as a whole: climate change"); and priority relative to other issues (e.g., agreement with "one must give priority to the fight against climate change without considering its negative impact on economic growth"). A complete list of these included survey-items may be found in Table SM1.

Survey-items relevant to perceptions of the benefits and costs of mitigation were identified as those asking whether measures to combat climate change (including specific measures, such as limiting fossil fuel imports, using energy more efficiently, or expanding renewable energy) will bring particular benefits (e.g., agreement with "fighting climate change can have a positive impact on the European economy"), those asking whether such measures will bring specified costs (e.g., agreement with "tackling climate change can harm our economy"), and those asking respondents to assess the balance of potential benefits and costs (e.g., "do you think that measures to combat climate change will have a negative, a positive, or neither a negative nor positive impact on your country's economic growth and job market?"). A complete list of these included survey-items can be found in Table SM2.

Survey-items on climate policy support were identified as those probing support for specific policies to combat climate change (e.g., "the EU should invest massively in renewable energies, such as wind and solar power"). All of these included survey-items are listed in Table SM3.

The climate survey-items excluded for lacking relevance fell into two categories. The first comprises survey-items asking about the cause of climate change (e.g., "is climate change caused by natural processes, human activity, or both?"). The second encompasses those questions that asked about individual behavior (e.g., "have you personally taken any action to fight climate change over the past six months?"), efficacy (e.g., "climate change is an unstoppable process, we cannot do anything about it"), interest (e.g., "in everyday life, we have to deal with many different issues, where we feel more or less interested. For each of the following, please indicate whether you are very interested, moderately interested, or not at all interested in: the environment and climate change"), subjective knowledge (e.g., "for each of these topics, can you tell me if you feel very well, fairly well, fairly badly or very badly informed? climate change"), or willingness to sacrifice (e.g., "would you be willing to give part of your income or pay more taxes, if you were sure that the extra money was used to: combat climate change?"). Although both of these groups of excluded questions are likely related to climate concern, an assessment of the benefits and costs of mitigation, and/or support for climate mitigation policies to some extent, they clearly represent distinct concepts.

The eligibility criterion was the number of countries covered by the survey item. Included survey-items were asked in ten or more countries; although ten is a small sample of countries, this is the size of the smallest survey analysed in Bush and Clayton (2023), and one strength of meta-analysis is its utility in combining insights from possibly underpowered studies. Excluded survey-items were asked in fewer than ten countries.

### A.2.1 Survey Items Used

Table SM1: Climate Change Concern Survey Items

| Survey | Survey Item Code | Question Text | Response Categories |
|---|---|---|---|
| amb2016 | future_problem4 | If nothing is done to reduce climate change in the future, how serious a problem do you think it will be for [country]? | 1 very serious / 2 somewhat serious / 3 not very serious / 4 not serious at all |
| amb2018 | future_problem4 | If nothing is done to reduce climate change in the future, how serious a problem do you think it will be for [country]? | 1 very serious / 2 somewhat serious / 3 not very serious / 4 not serious at all |
| amb2023 | future_problem4 | If nothing is done to reduce climate change in the future, how serious a problem do you think it will be for [country]? | 1 very serious / 2 somewhat serious / 3 not very serious / 4 not serious at all |
| arabb5 | problem4 | How serious a problem do you think the following issues are: climate change | 1 a very serious problem 2 a somewhat serious problem 3 not a very serious problem 4 not at all a serious problem |
| eb18 | concern4 | Finally, more generally, how concerned or worried are you about the following: the possible atmosphere damages affecting the world's weather brought about by the gas (carbon dioxide) emitted from burning coal and oil products | 1 a great deal / 2 a fair amount / 3 not very much / 4 not at all |
| eb29 | concern4 | Finally, more generally, how concerned or worried are you about the following: the possible atmosphere damages affecting the world's weather brought about by the gas (carbon dioxide) emitted from burning coal and oil products | 1 a great deal / 2 a fair amount / 3 not very much / 4 not at all |
| eb37 | concern4 | Are you very worried, somewhat worried, not very worried or not at all worried about the following problems? global warming | 1 very worried 2 somewhat worried 3 not very worried 4 not worried at all |
| eb37 | env_priority2a | In June 1992, the United Nations Conference for the Environment and Development will be held in Rio de Janeiro, Brazill. In your opinion, should this conference concern itself with the following problems, or not? Global warming | 1 yes should 2 no should not |
| eb431 | concern4 | Are you very worried, somewhat worried, not very worried or not at all worried about the following problems? global warming | 1 very worried 2 somewhat worried 3 not very worried 4 not worried at all |
| eb511 | concern4 | At present, are you very worried, somewhat worried, not very worried or not at all worried about the following? Global warming (greenhouse effect) | 1 very worried / 2 fairly worried / 3 not very worried / 4 not at all worried |
| eb511 | concern_trend3 | And, for each of these, would you say that you are more worried or less worried than 5 years ago? Global warming (greenhouse effect) | 1 more worried / 2 less worried / 3 neither more nor less worried |
| eb511 | harm_health3 | To what extent do you believe that the following could affect your health in the future? Climate change | 1 not at all / 2 a little / 3 a lot |
| eb57 | concern_urgent2 | Global warming and climate change are serious issues which need immediate action | 1 yes, it is the case / 2 no, it is not the case |
| eb58 | concern4 | At present, are you very worried, fairly worried, not very worried or not at all worried about the following topics? Climate change | 1 very worried / 2 fairly worried / 3 not very worried / 4 not at all worried |
| eb621 | env_definition2 | When people talk about "the environment", which of the following do you think of first? Climate change | 0 not mentioned / 1 mentioned |

Table SM1: Climate Change Concern Survey Items *(continued)*

| Survey | Survey Item Code | Question Text | Response Categories |
|---|---|---|---|
| eb621 | env_priority2 | From the following list, please list the five main environmental issues that you are worried about? Climate change | 0 not mentioned / 1 mentioned |
| eb672 | concern_urgent4 | Overall, thinking about global warming, do you think that it is a matter that should be dealt with by the European Union urgently or not? It should be dealt with by the European Union... | 1 very urgently / 2 fairly urgently / 3 not really urgently / 4 not at all urgently |
| eb682 | env_definition2 | When people talk about "the environment", which of the following do you think of first? Climate change | 0 not mentioned / 1 mentioned |
| eb682 | env_priority2 | From the following list, please list the five main environmental issues that you are worried about? Climate change | 0 not mentioned / 1 mentioned |
| eb692 | problem_priority2 | In your opinion, which of the following do you consider to be the most serious problem currently facing the world as a whole? Firstly? Climate change | 0 not mentioned / 1 mentioned |
| eb692 | concern10 | And how serious a problem do you think climate change is at this moment? | 1 not a serious problem at all / 23456789 / 10 an extremely serious problem |
| eb692 | denial4 | The seriousness of climate change has been exaggerated | 1 totally agree / 2 tend to agree / 3 tend to disagree / 4 totally disagree |
| eb711 | top3_challenge2 | In your opinion, which of the following do you consider to be the most serious problem currently facing the world as a whole: climate change | 0 not mentioned 1 mentioned |
| eb711 | concern10 | And how serious a problem do you think climate change is at this moment? | 1 not at all a serious problem / 23456789 / 10 an extremely serious problem |
| eb711 | denial4 | The seriousness of climate change has been exaggerated | 1 totally agree 2 tend to agree 3 tend to disagree 4 totally disagree |
| eb721 | top3_challenge2 | In your opinion, which of the following do you consider to be the most serious problem currently facing the world as a whole: climate change | 0 not mentioned 1 mentioned |
| eb721 | concern10 | And how serious a problem do you think climate change is at this moment? | 1 not at all a serious problem / 23456789 / 10 an extremely serious problem |
| eb721 | denial4 | The seriousness of climate change has been exaggerated | 1 totally agree 2 tend to agree 3 tend to disagree 4 totally disagree |
| eb742 | top2_country_challenge2 | What do you think are the two most important issues facing (OUR COUNTRY) at the moment? Climate change | 0 not mentioned 1 climate change |
| eb742 | top2_personal_challenge2 | What do you think are the two most important issues you are facing at the moment? Climate change | 0 not mentioned 1 climate change |
| eb742 | top2_eu_challenge2 | What do you think are the two most important issues facing the EU at the moment? Climate change | 0 not mentioned 1 climate change |
| eb753 | top2_country_challenge2 | What do you think are the two most important issues facing (OUR COUNTRY) at the moment? The environment, climate, and energy issues | 0 not mentioned 1 climate change |
| eb753 | top2_personal_challenge2 | What do you think are the two most important issues you are facing at the moment? Climate change | 0 not mentioned 1 climate change |
| eb753 | top2_eu_challenge2 | What do you think are the two most important issues facing the EU at the moment? The environment and climate change | 0 not mentioned 1 climate change |
| eb754 | top3_nat_security_challenge | What do you think are the most important challenges to the security of (NATIONALITY) citizens at the moment? Environmental issues/climate change | 0 not mentioned 1 mentioned |
| eb754 | top3_eu_security_challenge2 | And what do you think are the most important challenges to the security of EU citizens at the moment? Environmental issues/climate change | 0 not mentioned 1 mentioned |

Table SM1: Climate Change Concern Survey Items *(continued)*

| Survey | Survey Item Code | Question Text | Response Categories |
|---|---|---|---|
| eb754 | top3_challenge2 | In your opinion, which of the following do you consider to be the most serious problem currently facing the world as a whole: climate change | 0 not mentioned 1 mentioned |
| eb754 | concern10 | And how serious a problem do you think climate change is at this moment? | 1 not at all a serious problem / 23456789 / 10 an extremely serious problem |
| eb763 | top2_country_challenge2 | What do you think are the two most important issues facing (OUR COUNTRY) at the moment? The environment, climate, and energy issues | 0 not mentioned 1 climate change |
| eb763 | top2_personal_challenge2 | What do you think are the two most important issues you are facing at the moment? Climate change | 0 not mentioned 1 climate change |
| eb763 | top2_eu_challenge2 | What do you think are the two most important issues facing the EU at the moment? The environment and climate change | 0 not mentioned 1 climate change |
| eb773 | top2_country_challenge2 | What do you think are the two most important issues facing (OUR COUNTRY) at the moment? The environment, climate, and energy issues | 0 not mentioned 1 the environment and climate change |
| eb773 | top2_personal_challenge2 | What do you think are the two most important issues you are facing at the moment? The environment, climate, and energy issues | 0 not mentioned 1 the environment and climate change |
| eb773 | top2_eu_challenge2 | What do you think are the two most important issues facing the EU at the moment? The environment and climate change | 0 not mentioned 1 the environment and climate change |
| eb781 | top2_country_challenge2 | What do you think are the two most important issues facing (OUR COUNTRY) at the moment? The environment, climate, and energy issues | 0 not mentioned 1 the environment and climate change |
| eb781 | top2_personal_challenge2 | What do you think are the two most important issues you are facing at the moment? The environment, climate, and energy issues | 0 not mentioned 1 the environment and climate change |
| eb781 | top2_eu_challenge2 | What do you think are the two most important issues facing the EU at the moment? The environment and climate change | 0 not mentioned 1 the environment and climate change |
| eb793 | top2_country_challenge2 | What do you think are the two most important issues facing (OUR COUNTRY) at the moment? The environment, climate, and energy issues | 0 not mentioned 1 the environment and climate change |
| eb793 | top2_personal_challenge2 | What do you think are the two most important issues you are facing at the moment? The environment, climate, and energy issues | 0 not mentioned 1 the environment and climate change |
| eb793 | top2_eu_challenge2 | What do you think are the two most important issues facing the EU at the moment? The environment and climate change | 0 not mentioned 1 the environment and climate change |
| eb801 | top2_country_challenge2 | What do you think are the two most important issues facing (OUR COUNTRY) at the moment? The environment, climate, and energy issues | 0 not mentioned 1 the environment and climate change |
| eb801 | top2_personal_challenge2 | What do you think are the two most important issues you are facing at the moment? The environment, climate, and energy issues | 0 not mentioned 1 the environment and climate change |
| eb801 | top2_eu_challenge2 | What do you think are the two most important issues facing the EU at the moment? The environment and climate change | 0 not mentioned 1 the environment and climate change |
| eb802 | top3_challenge2 | Which of the following do you consider to be the single most serious problem facing the world as a whole? CLIMATE CHANGE (SUM) | 0 not mentioned 1 climate change |
| eb802 | concern10 | And how serious a problem do you think climate change is at this moment? | 1 not at all a serious problem / 23456789 / 10 an extremely serious problem |

Table SM1: Climate Change Concern Survey Items *(continued)*

| Survey | Survey Item Code | Question Text | Response Categories |
|---|---|---|---|
| eb812 | top2_country_challenge2 | What do you think are the two most important issues facing (OUR COUNTRY) at the moment? The environment, climate, and energy issues | 0 not mentioned 1 the environment, climate, and energy issues |
| eb812 | top2_personal_challenge2 | What do you think are the two most important issues you are facing at the moment? The environment, climate, and energy issues | 0 not mentioned 1 the environment, climate, and energy issues |
| eb812 | top2_eu_challenge2 | What do you think are the two most important issues facing the EU at the moment? The environment and climate change | 0 not mentioned 1 the environment and climate change |
| eb814 | top2_country_challenge2 | What do you think are the two most important issues facing (OUR COUNTRY) at the moment? The environment, climate, and energy issues | 0 not mentioned 1 the environment and climate change |
| eb814 | top2_personal_challenge2 | What do you think are the two most important issues you are facing at the moment? The environment, climate, and energy issues | 0 not mentioned 1 the environment and climate change |
| eb814 | top2_eu_challenge2 | What do you think are the two most important issues facing the EU at the moment? The environment and climate change | 0 not mentioned 1 the environment and climate change |
| eb823 | top2_country_challenge2 | What do you think are the two most important issues facing (OUR COUNTRY) at the moment? The environment, climate, and energy issues | 0 not mentioned 1 the environment and climate change |
| eb823 | top2_personal_challenge2 | What do you think are the two most important issues you are facing at the moment? The environment, climate, and energy issues | 0 not mentioned 1 the environment and climate change |
| eb823 | top2_eu_challenge2 | What do you think are the two most important issues facing the EU at the moment? The environment and climate change | 0 not mentioned 1 the environment and climate change |
| eb831 | top2_country_challenge2 | What do you think are the two most important issues facing (OUR COUNTRY) at the moment? The environment, climate, and energy issues | 0 not mentioned 1 the environment and climate change |
| eb831 | top2_personal_challenge2 | What do you think are the two most important issues you are facing at the moment? The environment, climate, and energy issues | 0 not mentioned 1 the environment and climate change |
| eb831 | top2_eu_challenge2 | What do you think are the two most important issues facing the EU at the moment? The environment and climate change | 0 not mentioned 1 the environment and climate change |
| eb833 | top2_country_challenge2 | What do you think are the two most important issues facing (OUR COUNTRY) at the moment? The environment, climate, and energy issues | 0 not mentioned 1 the environment and climate change |
| eb833 | top2_personal_challenge2 | What do you think are the two most important issues you are facing at the moment? The environment, climate, and energy issues | 0 not mentioned 1 the environment and climate change |
| eb833 | top2_eu_challenge2 | What do you think are the two most important issues facing the EU at the moment? The environment and climate change | 0 not mentioned 1 the environment and climate change |
| eb834 | top3_challenge2 | Which of the following do you consider to be the single most serious problem facing the world as a whole? CLIMATE CHANGE (SUM) | 0 not mentioned 1 climate change |
| eb834 | concern10 | And how serious a problem do you think climate change is at this moment? | 1 not at all a serious problem / 23456789 / 10 an extremely serious problem |
| eb843 | top2_country_challenge2 | What do you think are the two most important issues facing (OUR COUNTRY) at the moment? The environment, climate, and energy issues | 0 not mentioned 1 the environment and climate change |

Table SM1: Climate Change Concern Survey Items *(continued)*

| Survey | Survey Item Code | Question Text | Response Categories |
|---|---|---|---|
| eb843 | top2_personal_challenge2 | What do you think are the two most important issues you are facing at the moment? The environment, climate, and energy issues | 0 not mentioned 1 the environment and climate change |
| eb843 | top2_eu_challenge2 | What do you think are the two most important issues facing the EU at the moment? The environment and climate change | 0 not mentioned 1 the environment and climate change |
| eb852 | top2_country_challenge2 | What do you think are the two most important issues facing (OUR COUNTRY) at the moment? The environment, climate, and energy issues | 0 not mentioned 1 the environment and climate change |
| eb852 | top2_personal_challenge2 | What do you think are the two most important issues you are facing at the moment? The environment, climate, and energy issues | 0 not mentioned 1 the environment and climate change |
| eb852 | top2_eu_challenge2 | What do you think are the two most important issues facing the EU at the moment? The environment and climate change | 0 not mentioned 1 the environment and climate change |
| eb862 | top2_country_challenge2 | What do you think are the two most important issues facing (OUR COUNTRY) at the moment? The environment, climate, and energy issues | 0 not mentioned 1 the environment and climate change |
| eb862 | top2_personal_challenge2 | What do you think are the two most important issues you are facing at the moment? The environment, climate, and energy issues | 0 not mentioned 1 the environment and climate change |
| eb862 | top2_eu_challenge2 | What do you think are the two most important issues facing the EU at the moment? The environment and climate change | 0 not mentioned 1 the environment and climate change |
| eb871 | top3_challenge2 | Which of the following do you consider to be the most serious problem facing the world as a whole? climate change | 0 not mentioned 1 climate change |
| eb871 | concern10 | And how serious a problem do you think climate change is at this moment? | 1 not at all a serious problem / 23456789 / 10 an extremely serious problem |
| eb873 | top2_country_challenge2 | What do you think are the two most important issues facing (OUR COUNTRY) at the moment? The environment, climate, and energy issues | 0 not mentioned 1 the environment, climate, and energy issues |
| eb873 | top2_personal_challenge2 | What do you think are the two most important issues you are facing at the moment? The environment, climate, and energy issues | 0 not mentioned 1 the environment, climate, and energy issues |
| eb873 | top2_eu_challenge2 | What do you think are the two most important issues facing the EU at the moment? The environment and climate change | 0 not mentioned 1 climate change |
| eb883 | top2_country_challenge2 | What do you think are the two most important issues facing (OUR COUNTRY) at the moment? The environment, climate, and energy issues | 0 not mentioned 1 the environment and climate change |
| eb883 | top2_personal_challenge2 | What do you think are the two most important issues you are facing at the moment? The environment, climate, and energy issues | 0 not mentioned 1 the environment and climate change |
| eb883 | top2_eu_challenge2 | What do you think are the two most important issues facing the EU at the moment? The environment and climate change | 0 not mentioned 1 the environment and climate change |
| eb891 | top2_country_challenge2 | What do you think are the two most important issues facing (OUR COUNTRY) at the moment? The environment, climate, and energy issues | 0 not mentioned 1 the environment, climate, and energy issues |
| eb891 | top2_personal_challenge2 | What do you think are the two most important issues you are facing at the moment? The environment, climate, and energy issues | 0 not mentioned 1 the environment, climate, and energy issues |
| eb891 | top2_eu_challenge2 | What do you think are the two most important issues facing the EU at the moment? The environment and climate change | 0 not mentioned 1 climate change |

Table SM1: Climate Change Concern Survey Items *(continued)*

| Survey | Survey Item Code | Question Text | Response Categories |
|---|---|---|---|
| eb903 | top2_country_challenge2 | What do you think are the two most important issues facing (OUR COUNTRY) at the moment? The environment, climate, and energy issues | 0 not mentioned 1 the environment, climate, and energy issues |
| eb903 | top2_personal_challenge2 | What do you think are the two most important issues you are facing at the moment? The environment, climate, and energy issues | 0 not mentioned 1 the environment, climate, and energy issues |
| eb903 | top2_eu_challenge2 | What do you think are the two most important issues facing the EU at the moment? The environment and climate change | 0 not mentioned 1 climate change |
| eb913 | top3_challenge2 | Which of the following do you consider to be the single most serious problem facing the world as a whole? | 0 not mentioned 1 climate change |
| eb913 | concern10 | And how serious a problem do you think climate change is at this moment? | 1 not at all a serious problem / 23456789 / 10 an extremely serious problem |
| eb924 | concern_country10 | And how serious a problem do you think climate change is at this moment? In your country | 1 not at all a serious problem / 23456789 / 10 an extremely serious problem |
| eb924 | concern_eu10 | And how serious a problem do you think climate change is at this moment? In the EU | 1 not at all a serious problem / 23456789 / 10 an extremely serious problem |
| eb924 | top4_env_issue2 | From the following list, please pick the four environmental issues which you consider the most important. | 0 not mentioned 1 climate change |
| eb941 | top_eu_challenge2 | And which of the following do you think are the main challenges for the EU? | 0 not selected 1 environmental issues and climate change |
| eb941 | top2_challenge2 | Which two of the following should be prioritised in order to face major global challenges? | 0 not selected 1 environmental issues and climate change |
| eb943 | top2_country_challenge2 | What do you think are the two most important issues facing (OUR COUNTRY) at the moment? The environment and climate change | 0 not mentioned 1 the environment and climate change |
| eb943 | top2_personal_challenge2 | What do you think are the two most important issues you are facing at the moment? the environment and climate change | 0 not mentioned 1 the environment and climate change |
| eb943 | top2_eu_challenge2 | What do you think are the two most important issues facing (EU) at the moment? The environment and climate change | 0 not mentioned 1 the environment and climate change |
| eb951 | priority_eu2 | Which of the following topics would you like to see addressed in priority by the European Parliament? The environment and climate change | 0 not mentioned 1 the environment and climate change |
| eb951 | top3_challenge2 | Which of the following do you consider to be the single most serious problem facing the world as a whole? (First two) | 0 not mentioned 1 climate change |
| eb951 | concern10 | How serious a problem do you think climate change is at this moment? | 1 not at all a serious problem / 23456789 / 10 an extremely serious problem |
| eb953 | top2_country_challenge2 | What do you think are the two most important issues facing (OUR COUNTRY) at the moment? THE ENVIRONMENT & CLIMATE CHANGE | 0 not mentioned 1 the environment and climate change |
| eb953 | top2_personal_challenge2 | What do you think are the two most important issues facing (personal) at the moment?THE ENVIRONMENT & CLIMATE CHANGE | 0 not mentioned 1 the environment and climate change |
| eb953 | top2_eu_challenge2 | What do you think are the two most important issues facing (EU) at the moment?THE ENVIRONMENT & CLIMATE CHANGE | 0 not mentioned 1 the environment and climate change |
| eb961 | top2_priority2 | Which two of the following should be prioritised in order to face major global challenges? | 0 not mentioned 1 the environment and climate change |
| eb961 | top3_challenges_eu2 | And which of the following do you think are the main challenges for the EU? | 0 not mentioned 1 climate change |

Table SM1: Climate Change Concern Survey Items *(continued)*

| Survey | Survey Item Code | Question Text | Response Categories |
|---|---|---|---|
| eb963 | top2_country_challenge2 | What do you think are the two most important issues facing (OUR COUNTRY) at the moment? The environment and climate change | 0 not mentioned 1 the environment and climate change |
| eb963 | top2_personal_challenge2 | What do you think are the two most important issues that you are facing at the moment? The environment and climate change | 0 not mentioned 1 the environment and climate change |
| eb963 | top2_eu_challenge2 | What do you think are the two most important issues facing the EU at the moment? The environment and climate change | 0 not mentioned 1 the environment and climate change |
| eb975 | top2_country_challenge2 | What do you think are the two most important issues facing (OUR COUNTRY) at the moment? The environment and climate change | 0 not mentioned 1 the environment and climate change |
| eb975 | top2_personal_challenge2 | What do you think are the two most important issues that you are facing at the moment? The environment and climate change | 0 not mentioned 1 the environment and climate change |
| eb975 | top2_eu_challenge2 | What do you think are the two most important issues facing the EU at the moment? The environment and climate change | 0 not mentioned 1 the environment and climate change |
| eb981 | concern4 | To what extent are you currently worried or not about each of the following issues for your life and those of your close ones? Climate change | 1 very worried 2 slightly worried 3 not very worried 4 not at all worried |
| eb982 | top2_country_challenge2 | What do you think are the two most important issues facing (OUR COUNTRY) at the moment? The environment and climate change | 0 not mentioned 1 the environment and climate change |
| eb982 | top2_personal_challenge2 | What do you think are the two most important issues that you are facing at the moment? The environment and climate change | 0 not mentioned 1 the environment and climate change |
| eb982 | top2_eu_challenge2 | What do you think are the two most important issues facing the EU at the moment? The environment and climate change | 0 not mentioned 1 the environment and climate change |
| eb994 | top2_country_challenge2 | What do you think are the two most important issues facing (OUR COUNTRY) at the moment? The environment and climate change | 0 not mentioned 1 the environment and climate change |
| eb994 | top2_personal_challenge2 | What do you think are the two most important issues that you are facing at the moment? The environment and climate change | 0 not mentioned 1 the environment and climate change |
| eb994 | top2_eu_challenge2 | What do you think are the two most important issues facing the EU at the moment? The environment and climate change | 0 not mentioned 1 the environment and climate change |
| eb1002 | top2_country_challenge2 | What do you think are the two most important issues facing (OUR COUNTRY) at the moment? The environment and climate change | 0 not mentioned 1 the environment and climate change |
| eb1002 | top2_personal_challenge2 | What do you think are the two most important issues that you are facing at the moment? The environment and climate change | 0 not mentioned 1 the environment and climate change |
| eb1002 | top2_eu_challenge2 | What do you think are the two most important issues facing the EU at the moment? The environment and climate change | 0 not mentioned 1 the environment and climate change |
| eb1013 | top2_country_challenge2 | What do you think are the two most important issues facing (OUR COUNTRY) at the moment? The environment and climate change | 0 not mentioned 1 the environment and climate change |
| eb1013 | top2_personal_challenge2 | What do you think are the two most important issues that you are facing at the moment? The environment and climate change | 0 not mentioned 1 the environment and climate change |

Table SM1: Climate Change Concern Survey Items *(continued)*

| Survey | Survey Item Code | Question Text | Response Categories |
|---|---|---|---|
| eb1013 | top2_eu_challenge2 | What do you think are the two most important issues facing the EU at the moment? The environment and climate change | 0 not mentioned 1 the environment and climate change |
| eib2018 | concern5 | Thinking about climate change (e.g. global warming), which of the following descriptions apply best to yourself? | 1 alarmed (very worried) 2 concerned (worried) 3 neutral (not really worried) 4 doubtful (not sure climate change is happening) 5 denying (don't believe that climate change is happening) |
| eib2018 | when_threat6 | When will climate change become a threat to humanity–our lives and wellbeing? | 1 there is no climate change at all 2 it will become a threat in more than 20 years 3 it will become a threat in about 20 years 4 it will become a threat in about 10 years 5 it will become a threat in about 5 years 6 it is already a threat |
| eib2019 | top3_challenge2 | Climate change - In your opinion, what are the three biggest challenges that citizens in your country are currently facing? | 1 climate change 2 not mentioned |
| eib2019 | concern_personal3 | Do you feel that climate change is having an impact on your everyday life? | 1 yes, very much so 2 yes, somewhat 3 no, not at all |
| eib2019 | concern_children4 | Do you think that your children will feel the consequences of climate change in the future? | 1 yes, certainly 2 yes, probably 3 probably not 4 certainly not |
| eib2020 | top3_challenge2 | Climate change - In your opinion, what are the three biggest challenges that citizens in your country are currently facing? | 1 not mentioned 2 climate change |
| eib2020 | top_challenge2 | Do you think climate change and its consequences are the biggest challenges for humanity in the 21st century? | 1 no 2 yes |
| eib2020 | concern_personal4c | Do you feel that climate change is having an impact on your everyday life? | 1 no, not at all 2 no, not really 3 yes, somewhat 4 yes, very much so |
| eib2021 | top3_challenge2 | Climate change - In your opinion, what are the three biggest challenges that citizens in your country are currently facing? | 1 not mentioned 2 climate change |
| eib2021 | top_challenge2 | Do you think climate change and its consequences are the biggest challenges for humanity in the 21st century? | 1 no 2 yes |
| eib2021 | concern_personal4c | Do you feel that climate change is having an impact on your everyday life? | 1 no, not at all 2 no, not really 3 yes, somewhat 4 yes, very much so |
| eib2021 | concern_immigrate4 | Do you think you will have to move to another region or country in the future because of climate change | 1 certainly not 2 probably not 3 yes, probably 4 yes, certainly |
| eib2022 | top3_challenge2 | In your opinion, what are the three biggest challenges that people in your country are currently facing? Climate change | 1 not mentioned 2 climate change |
| eib2022 | concern_personal4c | Do you feel that climate change is having an impact on your everyday life? | 1 no, not at all 2 no, not really 3 yes, somewhat 4 yes, very much so |
| eib2022 | concern_catastrophe4 | If we don't drastically reduce our consumption of energy and goods in the coming years, we will be heading for a global catastrophe | 1 strongly agree 2 somewhat agree 3 somewhat disagree 4 strongly disagree |
| eib2022_afr | top5_challenges2 | In your opinion, what are the five biggest challenges that people in your country are currently facing? Climate change | 0 not mentioned 1 mentioned |
| eib2022_afr | concern_personal4c | Do you feel that climate change has an impact on your everyday life? | 1 yes, very much so 2 yes, somewhat 3 no, really 4 no, not at all |

Table SM1: Climate Change Concern Survey Items *(continued)*

| Survey | Survey Item Code | Question Text | Response Categories |
|---|---|---|---|
| eib2022_afr | concern_immigrate3 | Have you already had to move to another region or country because of climate change (for example, due a lack of access to water, change in precipitation patterns, etc.)? | 1 yes 2 no but I think i will have to do it in the future 3 no and i don't think it will be necessary in the future |
| eib2023 | top3_challenge2 | What are the three biggest challenges that people in your country are currently facing: climate change | 1 not mentioned 2 mentioned |
| eib2023_latam | top5_challenges2 | In your opinion, what are the five biggest challenges that people in your country are currently facing? Climate change | 1 not mentioned 2 mentioned |
| eib2023_latam | concern_personal4c | Do you feel that climate change is having an impact on your everyday life? | 1 no, not at all 2 no, not really 3 yes, somewhat 4 yes, very much so |
| eib2023_latam | concern_immigrate4 | Do you think you will have to move to another region or country in the future because of climate change | 1 certainly not 2 probably not 3 yes, certainly 4 yes, probably |
| eib2024 | top3_challenge2 | In your opinion, what are the three biggest challenges that people in your country are currently facing? Climate change | 1 climate change 2 no to climate change |
| eib2024 | concern_future4 | When you think of the potential impact of climate change on your life in the future, do you feel… | 1 very worried / 2 fairly worried / 3 not very worried / 4 not at all worried |
| eib2024 | concern_immigrate4a | Do you think you will have to take the following actions in the future because of climate change: move to a cooler region or country | 1 yes, certainly 2 yes, probably 3 probably not 4 certainly not |
| eib2024 | concern_immigrate4b | Do you think you will have to take the following actions in the future because of climate change: move to a less risky place, even locally (for example to avoid floods, forest fires or other extreme weather events) | 1 yes, certainly 2 yes, probably 3 probably not 4 certainly not |
| eib2024 | concern_property4 | How concerned are you that your home (house or apartment) or your car could be damaged in the coming years due to the impact of climate change ? | 1 very concerned 2 somewhat concerned 3 not very concerned 4 not concerned at all |
| ess8 | concern5 | How worried about climate change | 1 not at all worried 2 not very worried 3 somewhat worried 4 very worried 5 extremely worried |
| ess8 | impact11 | How good or bad do you think the impact of climate change will be on people across the world? | 0 extremely bad / 123456789 / 10 extremely good |
| ess8 | concern_energy5 | How worried, energy supply interrupted by natural disasters or extreme weather | 1 not at all worried 2 not very worried 3 somewhat worried 4 very worried 5 extremely worried |
| ess8 | denial4a | You may have heard the idea that the world's climate is changing due to increases in temperature over the past 100 years. What is your personal opinion on this? Do you think the world's climate is changing? | 1 definitely changing 2 probably changing 3 probably not changing 4 definitely not changing |
| ess10 | concern5 | How worried about climate change | 1 not at all worried 2 not very worried 3 somewhat worried 4 very worried 5 extremely worried |
| ess11 | concern5 | How worried about climate change | 1 not at all worried 2 not very worried 3 somewhat worried 4 very worried 5 extremely worried |
| feb123 | concern_future4 | How worried are you about future trends in these areas: climate change | 1 very much / 2 quite a lot / 3 not much / 4 not at all |
| issp1993 | danger5 | In general, do you think that a rise in the world's temperature caused by the 'greenhouse effect' is: | 1 extremely dangerous for the enviroment 2 very dangerous 3 somewhat dangerous 4 not very dangerous 5 not dangerous at all for the environment |

Table SM1: Climate Change Concern Survey Items *(continued)*

| Survey | Survey Item Code | Question Text | Response Categories |
|---|---|---|---|
| issp1993 | danger_personal5 | And do you think that a rise in the world's temperature caused by the 'greenhouse effect' is | 1 extremely dangerous for you and your family 2 very dangerous 3 somewhat dangerous 4 not very dangerous 5 not dangerous at all for you and your family |
| issp1993 | problem4a | Within the next 20 years or so, a rise in the world's temperature caused by the 'greenhouse effect' wil be one of the most serious problems for [country] | 1 definitely true 2 probably true 3 probably not true 4 definitely not true |
| issp2000 | danger5 | In general, do you think that a rise in the world's temperature caused by the 'greenhouse effect' is: | 1 extremely dangerous for the enviroment 2 very dangerous 3 somewhat dangerous 4 not very dangerous 5 not dangerous at all for the environment |
| issp2010 | danger5 | In general, do you think that a rise in the world's temperature caused by climate change is | 1 extremely dangerous for the enviroment 2 very dangerous 3 somewhat dangerous 4 not very dangerous 5 not dangerous at all for the environment |
| issp2020 | impact11 | How bad or good do you think the impacts of climate change will be for the world as a whole? | 0 extremely bad / 123456789 / 10 extremely good |
| issp2020 | impact_country11 | How bad or good do you think the impacts of climate change will be for [country]? | 0 extremely bad / 123456789 / 10 extremely good |
| issp2020 | danger5 | In general, do you think that a rise in the world's temperature caused by climate change is | 1 extremely dangerous for the enviroment 2 very dangerous 3 somewhat dangerous 4 not very dangerous 5 not dangerous at all for the environment |
| lb2007 | concern_country4 | How much do you think global warming affects (country)? | 1 a lot 2 some effect 3 little 4 no effect |
| lb2007 | concern_personal4b | And how much do you think it affects you? | 1 a lot 2 some effect 3 little 4 no effect |
| lb2007 | concern_economy4 | Please tell me how worried you are that environmental problems and global warming will negatively effect the country's development | 1 very worried 2 fairly worried 3 not very worried 4 not worried at all |
| lb2010 | concern_country4 | How much do you think global warming affects (country)? | 1 a lot 2 some effect 3 little 4 no effect |
| lb2010 | concern_personal4b | And how much do you think it affects you and your family? | 1 a lot 2 some effect 3 little 4 no effect |
| lb2011 | concern_country4 | How much do you think climate change affects (country)? | 1 a lot 2 some effect 3 little 4 no effect |
| lb2011 | concern_personal4b | And how much do you think it affects you and your family? | 1 a lot 2 some effect 3 little 4 no effect |
| lb2017 | denial4a | There is no such thing as the problem of climate change | 1 strongly agree 2 agree 3 disagree 4 strongly disagree |
| lb2017 | urgent4 | With which of the following statements about climate change do you agree the most? | 1 it is an urgent problem we need to address today 2 it is not yet an urgent problem, but it will be in the future 3 it is an urgent probleem but there is little to do, too late to act 4 it will never be necessary to address it |
| lits_wave2 | problem2 | In your opinion, which of the following do you consider to be a very serious problem currently facing the world: climate change | 0 not mentioned 1 mentioned |
| lits_wave2 | concern5 | How concerned are you about climate change? | 1 not at all / 234 / 5 extremely concerned |
| lits_wave2 | harm3 | As a result of climate change, do you think people in our country will be | 1 on balance, better-off 2 not affected much 3 on balance, worse-off |
| lits_wave2 | denial5 | The seriousness of climate change has been exaggerated | 1 strongly disagree 2 disagree 3 neither agree nor disagree 4 agree 5 strongly agree |

Table SM1: Climate Change Concern Survey Items *(continued)*

| Survey | Survey Item Code | Question Text | Response Categories |
|---|---|---|---|
| nsee_combo | denial4 | Scientists overstating evidence about global warming for their own interests | 1 strongly agree 2 somewhat agree 3 somewhat disagree 4 strongly disagree |
| nsee_combo | concern4 | How concerned are you about the issue of global warming | 1 very concerned 2 somewhat concerned 3 not too concerned 4 not at all concerned |
| nsee_combo | concern4 | How concerned are you about the issue of climate change | 1 very concerned 2 somewhat concerned 3 not too concerned 4 not at all concerned |
| pew2006 | concern5 | How much do you personally worry about global warming? | 0 never heard of it 1 not at all 2 only a little 3 a fair amount 4 a great deal |
| pew2007 | problem4 | On another topic, in your view, is global warming a very serious problem, somewhat serious, not too serious, or not a problem? | 1 very serious 2 somewhat serious 3 not too serious 4 not a problem |
| pew2008 | problem4 | On another topic, in your view, is global warming a very serious problem, somewhat serious, not too serious, or not a problem? | 1 very serious 2 somewhat serious 3 not too serious 4 not a problem |
| pew2009_spring | problem4 | On another topic, in your view, is global climate change a very serious problem, somewhat serious, not too serious, or not a problem? | 1 very serious 2 somewhat serious 3 not too serious 4 not a problem |
| pew2010 | problem4 | On another topic, in your view, is global climate change a very serious problem, somewhat serious, not too serious, or not a problem? | 1 very serious 2 somewhat serious 3 not too serious 4 not a problem |
| pew2013 | important_threat3b | I'd like your opinion about some possible international concerns for (survey country). Do you think that g. global climate change is a major threat, a minor threat or not a threat to (survey country)? | 1 major threat 2 minor threat 3 not a threat |
| pew2014_india | important_threat3b | I'd like your opinion about some possible international concerns for (survey country). Do you think that g. global climate change is a major threat, a minor threat or not a threat to (survey country)? | 1 major threat 2 minor threat 3 not a threat |
| pew2015 | concern4 | Please tell me how concerned you are, if at all, about: global climate change | 1 very concerned 2 somewhat concerned 3 not too concerned 4 not at all concerned |
| pew2015 | problem4 | On another topic, in your view, is global climate change a very serious problem, somewhat serious, not too serious, or not a problem? | 1 very serious 2 somewhat serious 3 not too serious 4 not a problem |
| pew2015 | when_harm4 | Do you think global climate change is harming people around the world now, will harm people in the next few years, will not harm people for many years or will never harm people? | 1 now 2 in the next few years 3 not for many years 4 never |
| pew2015 | concern_personal4 | How concerned are you, if at all, that global climate change will harm you personally at some point in your lifetime? | 1 very concerned 2 somewhat concerned 3 not too concerned 4 not at all concerned |
| pew2016 | important_threat3b | I'd like your opinion about some possible international concerns. Do you think that d. global climate change is a major threat, a minor threat or not a threat to (survey country)? | 1 major threat 2 minor threat 3 not a threat |
| pew2017 | important_threat3b | I'd like your opinion about some possible international concerns for (survey country). Do you think that d. global climate change is a major threat, a minor threat or not a threat to (survey country)? | 1 major threat 2 minor threat 3 not a threat |

Table SM1: Climate Change Concern Survey Items *(continued)*

| Survey | Survey Item Code | Question Text | Response Categories |
|---|---|---|---|
| pew2018 | important_threat3b | I'd like your opinion about some possible international concerns for (survey country). Do you think that d. global climate change is a major threat, a minor threat or not a threat to (survey country)? | 1 major threat 2 minor threat 3 not a threat |
| pew2020 | important_threat3b | I'd like your opinion about some possible international concerns for (survey country). Do you think that d. global climate change is a major threat, a minor threat or not a threat to (survey country)? | 1 major threat 2 minor threat 3 not a threat |
| pew2021 | concern_personal4 | How concerned are you, if at all, that global climate change will harm you personally at some point in your lifetime? | 1 very concerned 2 somewhat concerned 3 not too concerned 4 not at all concerned |
| pipa2007 | important_threat3a | Below is a list of possible threats to the vital interest of [survey country] in the next 10 years. For each one, please select whether you see this as a critical threat, an important but not critical threat, or not an important threat at all...Global Warming | 1 critical threat 2 important but not critical threat 3 not important threat |
| pipa2007 | urgent3 | There is a controversy over what the countries of the world, including [survey country], should do about the problem of global warming. Here are three statements. Please tell me which statement comes closest to your own point of view." | 1 until we are sure that global warming is really a problem, we should not take any steps that would have economic costs 2 the problem of global warming should be addressed, but its effects will be gradual, so we can deal with the problem gradually by taking steps that are low in cost 3 global warming is a serious and pressing problem. we should begin taking steps now even if this involves significant costs |
| pipa2009 | problem4 | In your view, is climate change, also called global warming, a serious problem? | 1 serious problem 2 somewhat serious 3 not too serious 4 not a problem |
| pipa2009 | harm_food4 | If climate change is left unchecked worldwide, how much do you think climate change, will affect each of the following in our country: The types of food we produce | 1 none at all 2 not very much 3 some 4 a lot |
| pipa2009 | harm_wildlife4 | If climate change is left unchecked worldwide, how much do you think climate change, will affect each of the following in our country: The types of plants and animals that can live here | 1 none at all 2 not very much 3 some 4 a lot |
| pipa2009 | harm_water4 | If climate change is left unchecked worldwide, how much do you think climate change, will affect each of the following in our country: Rainfall and available water resources | 1 none at all 2 not very much 3 some 4 a lot |
| pipa2009 | harm_prices4 | If climate change is left unchecked worldwide, how much do you think climate change, will affect each of the following in our country: The price of food and other essential goods | 1 none at all 2 not very much 3 some 4 a lot |
| pipa2009 | harm_disasters4 | If climate change is left unchecked worldwide, how much do you think climate change, will affect each of the following in our country: The likelihood of natural disasters, like droughts or floods | 1 none at all 2 not very much 3 some 4 a lot |
| pipa2009 | harm_coastline4 | If climate change is left unchecked worldwide, how much do you think climate change, will affect each of the following in our country: our coastline | 1 none at all 2 not very much 3 some 4 a lot |

Table SM1: Climate Change Concern Survey Items *(continued)*

| Survey | Survey Item Code | Question Text | Response Categories |
|---|---|---|---|
| pipa2009 | harm_homes4 | If climate change is left unchecked worldwide, how much do you think climate change, will affect each of the following in our country: People's need to move their homes to different locations | 1 none at all 2 not very much 3 some 4 a lot |
| pipa2009 | when_harm6 | When do you think climate change will start to substantially harm people in [country]? | 1 people are being harmed now 2 in 10 years 3 in 25 years 4 in 50 years 5 in 100 years 6 never |
| pipa201007 | problem4 | In your view, is climate change, also called global warming, a serious problem? | 1 serious problem 2 somewhat serious 3 not too serious 4 not a problem |
| pipa201007 | harm_food4 | If climate change is left unchecked worldwide, how much do you think climate change, will affect each of the following in our country: The types of food we produce | 1 none at all 2 not very much 3 some 4 a lot |
| pipa201007 | harm_wildlife4 | If climate change is left unchecked worldwide, how much do you think climate change, will affect each of the following in our country: The types of plants and animals that can live here | 1 none at all 2 not very much 3 some 4 a lot |
| pipa201007 | harm_water4 | If climate change is left unchecked worldwide, how much do you think climate change, will affect each of the following in our country: Rainfall and available water resources | 1 none at all 2 not very much 3 some 4 a lot |
| pipa201007 | harm_prices4 | If climate change is left unchecked worldwide, how much do you think climate change, will affect each of the following in our country: The price of food and other essential goods | 1 none at all 2 not very much 3 some 4 a lot |
| pipa201007 | harm_disasters4 | If climate change is left unchecked worldwide, how much do you think climate change, will affect each of the following in our country: The likelihood of natural disasters, like droughts or floods | 1 none at all 2 not very much 3 some 4 a lot |
| pipa201007 | harm_coastline4 | If climate change is left unchecked worldwide, how much do you think climate change, will affect each of the following in our country: our coastline | 1 none at all 2 not very much 3 some 4 a lot |
| pipa201007 | harm_homes4 | If climate change is left unchecked worldwide, how much do you think climate change, will affect each of the following in our country: People's need to move their homes to different locations | 1 none at all 2 not very much 3 some 4 a lot |
| pipa201007 | when_harm6 | When do you think climate change will start to substantially harm people in [country]? | 1 people are being harmed now 2 in 10 years 3 in 25 years 4 in 50 years 5 in 100 years 6 never |
| rtm2022 | concern_consequences4 | Given current events, how concerned are you about each of the following in [country]? Consequences of climate change | 1 not at all concerned / 2 not so concerned / 3 somewhat concerned / 4 very concerned |
| rtm2022 | concern4 | How worried are you about climate change? | 1 not at all concerned / 2 not very concerned / 3 somewhat concerned / 4 very concerned |
| szazadveg_cc20 | concern5 | How worried are you about climate change? | 1 not at all worried / 2 not very worried / 3 somewhat worried / 4 very worried / 5 extremely worried |
| tt2002 | important_threat3a | I am going to read you a list of possible threats to the vital interest of [own country] in the next 10 years. For each one, please tell if you see this as an extremely important threat, an important threat, or not an important threat at all | 1 extremely important threat 2 important threat 3 not important threat |

Table SM1: Climate Change Concern Survey Items *(continued)*

| Survey | Survey Item Code | Question Text | Response Categories |
|---|---|---|---|
| tt2002 | important_threat3 | I am going to read you a list of possible international threats to Europe/US in the next 10 years. Please tell me if you think each one on the list is an extremely important threat, an important threat, or not an important threat at all. f) the effects of global warming | 1 extremely important threat 2 important threat 3 not important threat |
| tt2005 | important_threat3 | I am going to read you a list of possible international threats to Europe/US in the next 10 years. Please tell me if you think each one on the list is an extremely important threat, an important threat, or not an important threat at all. f) the effects of global warming | 1 extremely important threat 2 important threat 3 not important threat |
| tt2005 | concern_personal4a | And in the next 10 years, please tell me how likely you are to be personally affected by each of the following threats: The effects of global warming | 1 very likely 2 somewhat likely 3 not too likely 4 not likely at all |
| tt2006 | important_threat3 | I am going to read you a list of possible international threats to Europe/US in the next 10 years. Please tell me if you think each one on the list is an extremely important threat, an important threat, or not an important threat at all. f) the effects of global warming | 1 extremely important threat 2 important threat 3 not important threat |
| tt2006 | important_threat4 | I am going to read you a list of possible international threats to Europe/US in the next 10 years. Please tell me if you think each one on the list is an extremely important threat, an important threat, or not an important threat at all. f) the effects of global warming | 1 extremely important threat 2 important threat 3 not important threat |
| tt2007 | concern_personal4a | And in the next 10 years, please tell me how likely you are to be personally affected by each of the following threats: The effects of global warming | 1 very likely 2 somewhat likely 3 not too likely 4 not likely at all |
| tt2008 | concern_personal4a | And in the next 10 years, please tell me how likely you are to be personally affected by each of the following threats. - The effects of global warming | 1 very likely 2 somewhat likely 3 not too likely 4 not likely at all |
| tt2009 | concern4 | To what extent are you concerned about: climate change | 1 very concerned 2 somewhat concerned 3 not very concerned 4 not at all concerned |
| wrp19_21 | threat3 | Will climate change be a threat to people in country in next 20 years? | 1 very serious threat / 2 somewhat serious threat / 3 not a threat at all |
| wvs_combo | concern_global4 | Now let's consider environmental problems in the world as a whole. Please, tell me how serious you consider each of the following to be for the world as a whole. Is it very serious, somewhat serious, not very serious or not serious at all? Global warming or the greenhouse effect | 1 very serious / 2 somewhat serious / 3 not very serious / 4 not serious at all |

* Survey dataset codes correspond to those used in the DCPOtools R package (Solt, Hu, and Tai 2019).

Table SM2: Benefits and Costs of Climate Change Mitigation Survey Items

| Survey | Survey Item Code | Question Text | Response Categories |
|---|---|---|---|
| eb1002 | value_renewables4 | In the long run, renewable energy can limit the price we pay for our energy consumption | 1 totally agree 2 tend to agree 3 tend to disagree 4 totally disagree |
| eb1002 | reduce_ff_imports4c | Reducing imports of oil and gas and investing in renewable energy is important for our overall security | 1 totally agree 2 tend to agree 3 tend to disagree 4 totally disagree |
| eb1013 | value_renewables4 | In the long run, renewable energy can limit the price we pay for our energy consumption | 1 totally agree 2 tend to agree 3 tend to disagree 4 totally disagree |
| eb1013 | reduce_ff_imports4c | Reducing imports of oil and gas and investing in renewable energy is important for our overall security | 1 totally agree 2 tend to agree 3 tend to disagree 4 totally disagree |
| eb692 | mitigation_growth4 | Fighting climate change can have a positive impact on the European economy | 1 totally agree 2 tend to agree 3 tend to disagree 4 totally disagree |
| eb711 | mitigation_growth4 | Fighting climate change can have a positive impact on the European economy | 1 totally agree 2 tend to agree 3 tend to disagree 4 totally disagree |
| eb721 | mitigation_growth4 | Fighting climate change can have a positive impact on the European economy | 1 totally agree 2 tend to agree 3 tend to disagree 4 totally disagree |
| eb754 | mitigation_jobs4 | Fighting climate change and using energy more efficiently can create economic growth and jobs in the EU | 1 totally agree 2 tend to agree 3 tend to disagree 4 totally disagree |
| eb802 | mitigation_jobs4 | Fighting climate change and using energy more efficiently can create economic growth and jobs in the EU | 1 totally agree 2 tend to agree 3 tend to disagree 4 totally disagree |
| eb802 | reduce_ff_imports4a | Reducing fossil fuel imports from outside the EU can benefit the EU economically | 1 totally agree 2 tend to agree 3 tend to disagree 4 totally disagree |
| eb834 | mitigation_jobs4 | Fighting climate change and using energy more efficiently can create economic growth and jobs in the EU | 1 totally agree 2 tend to agree 3 tend to disagree 4 totally disagree |
| eb834 | tech_jobs4 | Promoting EU expertise in clean technologies to countries outside the EU can help create new jobs in the EU | 1 totally agree 2 tend to agree 3 tend to disagree 4 totally disagree |
| eb834 | reduce_ff_imports4a | Reducing fossil fuel imports from outside the EU can benefit the EU economically | 1 totally agree 2 tend to agree 3 tend to disagree 4 totally disagree |
| eb834 | reduce_ff_imports4b | Reducing fossil fuel imports from outside the EU can increase energy security | 1 totally agree 2 tend to agree 3 tend to disagree 4 totally disagree |
| eb871 | mitigation_jobs4 | Fighting climate change and using energy more efficiently can create economic growth and jobs in the EU | 1 totally agree 2 tend to agree 3 tend to disagree 4 totally disagree |
| eb871 | tech_jobs4 | Promoting EU expertise in clean technologies to countries outside the EU can help create new jobs in the EU | 1 totally agree 2 tend to agree 3 tend to disagree 4 totally disagree |
| eb871 | reduce_ff_imports4a | Reducing fossil fuel imports from outside the EU can benefit the EU economically | 1 totally agree 2 tend to agree 3 tend to disagree 4 totally disagree |
| eb871 | reduce_ff_imports4b | Reducing fossil fuel imports from outside the EU can increase energy security | 1 totally agree 2 tend to agree 3 tend to disagree 4 totally disagree |
| eb902 | mitigation_jobs4 | Fighting climate change and using energy more efficiently can create economic growth and jobs in the EU | 1 totally agree 2 tend to agree 3 tend to disagree 4 totally disagree |
| eb902 | tech_jobs4 | Promoting EU expertise in clean technologies to countries outside the EU can help create new jobs in the EU | 1 totally agree 2 tend to agree 3 tend to disagree 4 totally disagree |
| eb902 | reduce_ff_imports4a | Reducing fossil fuel imports from outside the EU can benefit the EU economically | 1 totally agree 2 tend to agree 3 tend to disagree 4 totally disagree |
| eb902 | reduce_ff_imports4b | Reducing fossil fuel imports from outside the EU can increase energy security | 1 totally agree 2 tend to agree 3 tend to disagree 4 totally disagree |
| eb902 | support_renewables4 | More public financial support should be given to the transition to clean energies even if it means subsidies to fossil fuels should be reduced | 1 totally agree 2 tend to agree 3 tend to disagree 4 totally disagree |

Table SM2: Benefits and Costs of Climate Change Mitigation Survey Items *(continued)*

| Survey | Survey Item Code | Question Text | Response Categories |
|---|---|---|---|
| eb913 | tech_jobs4 | Promoting EU expertise in clean technologies to countries outside the EU can help create new jobs in the EU | 1 totally agree 2 tend to agree 3 tend to disagree 4 totally disagree |
| eb913 | reduce_ff_imports4 | Reducing fossil fuel imports from outside the EU can increase energy security and benefit the EU economically | 1 totally agree 2 tend to agree 3 tend to disagree 4 totally disagree |
| eb913 | innovate4 | Taking action on climate change will lead to innovation that will make EU companies more competitive | 1 totally agree 2 tend to agree 3 tend to disagree 4 totally disagree |
| eb913 | support_renewables4 | More public financial support should be given to the transition to clean energies even if it means subsidies to fossil fuels should be reduced | 1 totally agree 2 tend to agree 3 tend to disagree 4 totally disagree |
| eb913 | mitigation_positive4 | Adapting to the adverse impacts of climate change can have positive outcomes for citizens in the EU | 1 totally agree 2 tend to agree 3 tend to disagree 4 totally disagree |
| eb951 | tech_jobs4 | Promoting EU expertise in clean technologies to countries outside the EU can help create new jobs in the EU | 1 totally agree 2 tend to agree 3 tend to disagree 4 totally disagree |
| eb951 | mitigation_health4a | Tackling climate change and environmental issues should be a priority to improve public health | 1 totally agree 2 tend to agree 3 tend to disagree 4 totally disagree |
| eb951 | mitigation_save4a | The costs of the damages due to climate change are much higher than the costs of the investments needed for a green transition | 1 totally agree 2 tend to agree 3 tend to disagree 4 totally disagree |
| eb951 | reduce_ff_imports4 | Reducing fossil fuel imports from outside the EU can increase energy security and benefit the EU economically | 1 totally agree 2 tend to agree 3 tend to disagree 4 totally disagree |
| eb951 | innovate4 | Taking action on climate change will lead to innovation that will make EU companies more competitive | 1 totally agree 2 tend to agree 3 tend to disagree 4 totally disagree |
| eb951 | mitigation_positive4 | Adapting to the adverse impacts of climate change can have positive outcomes for citizens in the EU | 1 totally agree 2 tend to agree 3 tend to disagree 4 totally disagree |
| eb961 | mitigation_jobs4 | Tackling climate change can create new opportunities for innovation, investment and jobs | 1 totally agree 2 tend to agree 3 tend to disagree 4 totally disagree |
| eb961 | mitigation_health4 | Tackling climate change can help improve your own health and well-being | 1 totally agree 2 tend to agree 3 tend to disagree 4 totally disagree |
| eb961 | mitigation_harm4 | Tackling climate change can harm our economy | 1 totally agree 2 tend to agree 3 tend to disagree 4 totally disagree |
| eb961 | mitigation_save4 | Tackling climate change now can help to reduce the costs of greater ecological damage in the future | 1 totally agree 2 tend to agree 3 tend to disagree 4 totally disagree |
| eb973 | mitigation_jobs4 | Tackling climate change can create new opportunities for innovation, investment and jobs | 1 totally agree 2 tend to agree 3 tend to disagree 4 totally disagree |
| eb973 | mitigation_health4 | Tackling climate change can help improve your own health and well-being | 1 totally agree 2 tend to agree 3 tend to disagree 4 totally disagree |
| eb973 | mitigation_harm4 | Tackling climate change can harm our economy | 1 totally agree 2 tend to agree 3 tend to disagree 4 totally disagree |
| eb973 | mitigation_save4 | Tackling climate change now can help to reduce the costs of greater ecological damage in the future | 1 totally agree 2 tend to agree 3 tend to disagree 4 totally disagree |
| eib2018 | mitigation_jobs5 | Do you think that measures to combat climate change will have a negative, a positive, or neither a negative nor positive impact on your country's economic growth and job market? | 1 very negative 2 rather negative 3 neither negative nor positive 4 rather positive 5 very positive |
| eib2021 | mitigation_jobs2 | Do you believe that policies to fight climate change will… | 1 create more jobs than they will remove existing ones 2 remove more jobs than they will create new ones |

Table SM2: Benefits and Costs of Climate Change Mitigation Survey Items *(continued)*

| Survey | Survey Item Code | Question Text | Response Categories |
|---|---|---|---|
| eib2021 | mitigation_living2 | Do you believe that policies to fight climate change will… | 1 improve your living standards (daily life convenience, food, health) 2 worsen your living standards (daily life convenience, food, health) |
| eib2021 | mitigation_growth2 | Do you believe that policies to fight climate change will… | 1 be a source of economic growth and wealth for your country 2 slow or reduce economic growth in your country |
| eib2021 | mitigation_cost2 | Do you believe that policies to fight climate change will… | 1 reduce your purchasing power 2 increase your purchasing power |
| eib2021 | burden_unemployment4 | Are you concerned that you will lose your job in the future because it will become incompatible with the fight against climate change? | 1 certainly not 2 probably not 3 yes, probably 4 yes, certainly |
| eib2023 | mitigation_living2 | What impact do you think the measures adopted by your country to fight climate change and protect the environment will have? | 1 they will negatively affect my daily life, including the quality of food and my health 2 they will improve my daily life quality, including the quality fo food and my health |
| eib2023 | mitigation_jobs2 | What impact do you think the measures adopted by your country to fight climate change and protect the environment will have? | 1 they will destroy more jobs than they will create new ones 2 they will creat more jobs than they destroy existing ones |
| eib2023 | mitigation_cost2 | What impact do you think the measures adopted by your country to fight climate change and protect the environment will have? | 1 they will reduce my purchasing power 2 they will increase my purchasing power |
| eib2023_latam | mitigation_jobs2 | What impact do you think that policies to fight climate change and protect the environment in your country will have? | 1 they will create more new jobs than they remove existing ones 2 they will remove more jobs than they create new ones |
| eib2023_latam | mitigation_living2 | What impact do you think that policies to fight climate change and protect the environment in your country will have? | 1 they will worsen your daily life convenience, food, and health 2 they will improve your dail life convenience, food, and health |
| eib2023_latam | mitigation_growth2 | What impact do you think that policies to fight climate change and protect the environment in your country will have? | 1 they will slow down or reduce economic growth in your country 2 they will be a source of economic growth and wealth for your country |
| eib2024 | mitigation_value4 | Climate change adaptation requires costs now to avoid greater costs in the future | 1 strongly agree 2 somewhat agree 3 somewhat disagree 4 strongly disagree |
| eib2024 | mitigation_jobs4a | Investing in climate adaption infrastructure can help create jobs and boost the local economy | 1 strongly agree 2 somewhat agree 3 somewhat disagree 4 strongly disagree |
| rtm2022 | burden_energy_cost4 | The government can take a number of environmental regulatory steps to reduce [country]'s contribution to climate change, such as building up green infrastructure, emissions limits and carbon taxes. But these environmental measures to mitigate climate change can have social and economic consequences on people today. Keeping in mind the effects of different environmental policies to combat climate change, what degree are you concerned about the following possible economic outcomes in [country]? Rising energy/fuel costs | 1 not at all concerned / 2 not so concerned / 3 somewhat concerned / 4 very concerned |

Table SM2: Benefits and Costs of Climate Change Mitigation Survey Items *(continued)*

| Survey | Survey Item Code | Question Text | Response Categories |
|---|---|---|---|
| rtm2022 | burden_inflation4 | The government can take a number of environmental regulatory steps to reduce [country]'s contribution to climate change, such as building up green infrastructure, emissions limits and carbon taxes. But these environmental measures to mitigate climate change can have social and economic consequences on people today. Keeping in mind the effects of different environmental policies to combat climate change, what degree are you concerned about the following possible economic outcomes in [country]? Rising costs of food and other goods | 1 not at all concerned / 2 not so concerned / 3 somewhat concerned / 4 very concerned |
| rtm2022 | burden_job_loss4 | The government can take a number of environmental regulatory steps to reduce [country]'s contribution to climate change, such as building up green infrastructure, emissions limits and carbon taxes. But these environmental measures to mitigate climate change can have social and economic consequences on people today. Keeping in mind the effects of different environmental policies to combat climate change, what degree are you concerned about the following possible economic outcomes in [country]? A loss of jobs in industries that have negative environmental impacts, such as coal mining and oil extraction | 1 not at all concerned / 2 not so concerned / 3 somewhat concerned / 4 very concerned |
| rtm2022 | burden_skill_shortage4 | The government can take a number of environmental regulatory steps to reduce [country]'s contribution to climate change, such as building up green infrastructure, emissions limits and carbon taxes. But these environmental measures to mitigate climate change can have social and economic consequences on people today. Keeping in mind the effects of different environmental policies to combat climate change, what degree are you concerned about the following possible economic outcomes in [country]? Having enough skilled workers to fill demand for green jobs | 1 not at all concerned / 2 not so concerned / 3 somewhat concerned / 4 very concerned |
| rtm2022 | burden_housing4 | The government can take a number of environmental regulatory steps to reduce [country]'s contribution to climate change, such as building up green infrastructure, emissions limits and carbon taxes. But these environmental measures to mitigate climate change can have social and economic consequences on people today. Keeping in mind the effects of different environmental policies to combat climate change, what degree are you concerned about the following possible economic outcomes in [country]? Housing relocation away from environmentally degraded spaces | 1 not at all concerned / 2 not so concerned / 3 somewhat concerned / 4 very concerned |

Table SM2: Benefits and Costs of Climate Change Mitigation Survey Items *(continued)*

| Survey | Survey Item Code | Question Text | Response Categories |
|---|---|---|---|
| rtm2022 | burden_growth4 | The government can take a number of environmental regulatory steps to reduce [country]'s contribution to climate change, such as building up green infrastructure, emissions limits and carbon taxes. But these environmental measures to mitigate climate change can have social and economic consequences on people today. Keeping in mind the effects of different environmental policies to combat climate change, what degree are you concerned about the following possible economic outcomes in [country]? Lower economic growth | 1 not at all concerned / 2 not so concerned / 3 somewhat concerned / 4 very concerned |
| rtm2022 | burden_hvac4 | The government can take a number of environmental regulatory steps to reduce [country]'s contribution to climate change, such as building up green infrastructure, emissions limits and carbon taxes. But these environmental measures to mitigate climate change can have social and economic consequences on people today. Keeping in mind the effects of different environmental policies to combat climate change, what degree are you concerned about the following possible economic outcomes in [country]? Costs of (mandatory) climate-neutral adaptation of heating and cooling systems (e.g. renewable energy systems, energy efficiency renovations) | 1 not at all concerned / 2 not so concerned / 3 somewhat concerned / 4 very concerned |

* Survey dataset codes correspond to those used in the DCPOtools R package (Solt, Hu, and Tai 2019).

Table SM3: Climate Change Policy Support Survey Items

| Survey | Survey Item Code | Question Text | Response Categories |
|---|---|---|---|
| eb1002 | invest_renewables4 | The EU should invest massively in renewable energies, such as wind and solar power | 1 totally agree 2 tend to agree 3 tend to disagree 4 totally disagree |
| eb1013 | invest_renewables4 | The EU should invest massively in renewable energies, such as wind and solar power | 1 totally agree 2 tend to agree 3 tend to disagree 4 totally disagree |
| eb672 | policy_carbon4a | More precisely, please tell me to what extent do you agree or disagree with the following statement about climate change: The European Union should urgently put new policies in place to reduce greenhouse gas emission by at least 20% by 2020 | 1 totally agree / 2 tend to agree / 3 tend to disagree / 4 totally disagree |
| eb692 | biofuel4 | Alternative fuels, such as "bio fuels", should be used to reduce greenhouse gas emissions | 1 totally agree / 2 tend to agree / 3 tend to disagree / 4 totally disagree |
| eb692 | policy_emissions3 | The European Union has the objective of reducing its greenhouse gas emissions by at least 20% by 2020 compared to 1990. Thinking about this objective, would you say that it is too ambitious, about right or too modest? | 1 too ambitious / 2 about right / 3 too modest |
| eb692 | policy_emissions3a | In order to limit the impact of climate change, the EU is also proposing an international agreement which would commit the major world economies to a target of 30% reduction of greenhouse gas emissions by 2020 compared to 1990. Thinking about this objective, would you say that it is too ambitious, about right or too modest? | 1 too ambitious / 2 about right / 3 too modest |
| eb692 | policy_mandate3 | The European Union has the objective of increasing the share of renewable energy to 20% by 2020. Thinking about this objective, would you say that it is too ambitious, about right or too modest? | 1 too ambitious / 2 about right / 3 too modest |
| eb711 | alternate_fuels4a | Alternative fuels, such as "bio fuels", should be used to reduce greenhouse gas emissions | 1 totally agree 2 tend to agree 3 tend to disagree 4 totally disagree |
| eb721 | alternate_fuels4a | Alternative fuels, such as "bio fuels", should be used to reduce greenhouse gas emissions | 1 totally in favor 2 fairly in favor 3 fairly opposed 4 totally opposed |
| eb721 | alternate_fuels4b | Alternative fuels should be used to reduce greenhouse gas emissions | 1 totally in favor 2 fairly in favor 3 fairly opposed 4 totally opposed |
| eb742 | refocus_research2 | Thinking about innovation, what do you think should be the EU's priority? Refocus research on new challenges such as cllimate change, energy and resource efficiency | 0 not mentioned 1 mentioned |
| eb751 | favor_solar4 | To what extent are you in favour of or opposed to the use of the following sources of energy in (OUR COUNTRY)? Solar energy | 1 totally in favor 2 fairly in favor 3 fairly opposed 4 totally opposed |
| eb751 | favor_wind4 | To what extent are you in favour of or opposed to the use of the following sources of energy in (OUR COUNTRY)? Wind energy | 1 totally in favor 2 fairly in favor 3 fairly opposed 4 totally opposed |
| eb751 | favor_hydro4 | To what extent are you in favour of or opposed to the use of the following sources of energy in (OUR COUNTRY)? Hydroelectric energy | 1 totally in favor 2 fairly in favor 3 fairly opposed 4 totally opposed |
| eb751 | favor_biomass4 | To what extent are you in favour of or opposed to the use of the following sources of energy in (OUR COUNTRY)? Biomass energy | 1 totally in favor 2 fairly in favor 3 fairly opposed 4 totally opposed |
| eb751 | favor_coal4 | To what extent are you in favour of or opposed to the use of the following sources of energy in (OUR COUNTRY)? Coal | 1 totally in favor 2 fairly in favor 3 fairly opposed 4 totally opposed |

Table SM3: Climate Change Policy Support Survey Items *(continued)*

| Survey | Survey Item Code | Question Text | Response Categories |
|---|---|---|---|
| eb751 | favor_natural_gas4 | To what extent are you in favour of or opposed to the use of the following sources of energy in (OUR COUNTRY)? Natural gas | 1 totally in favor 2 fairly in favor 3 fairly opposed 4 totally opposed |
| eb751 | favor_nuclear4 | To what extent are you in favour of or opposed to the use of the following sources of energy in (OUR COUNTRY)? Nuclear energy | 1 totally in favor 2 fairly in favor 3 fairly opposed 4 totally opposed |
| eb753 | refocus_research2 | Thinking about innovation, what do you think should be the EU's priority? Refocus research on new challenges such as cllimate change, energy and resource efficiency | 0 not mentioned 1 mentioned |
| eb753 | top3_eu_industry_priority2 | Thinking about industry, what do you think should be the EU's priority? To make the most of the opportunities of the green economy (i.e. environmental technologies, recycling, renewable energy) | 0 not mentioned 1 mentioned |
| eb753 | top3_eu_budget_priority2 | And on which of the following would you like EU budget to be spent? Climate change and environmental protection | 0 not mentioned 1 mentioned |
| eb754 | tax_energy4 | Taxation should be based more on the way we use energy | 1 totally agree 2 tend to agree 3 tend to disagree 4 totally disagree |
| eb763 | refocus_research2 | Thinking about innovation, what do you think should be the EU's priority? Refocus research on new challenges such as cllimate change, energy and resource efficiency | 0 not mentioned 1 mentioned |
| eb802 | important_renewables4 | How important do you think it is that the (OUR NATIONALITY) government sets targets to increase the amount of renewable energyused, such as wind or solar power, by 2030? | 1 very important 2 fairly important 3 not very important 4 not at all important |
| eb802 | important_reduce4 | How important do you think it is that the (OUR NATIONALITY) government provides support for improving energy efficiency (for example, by encouraging people to insulate their home or purchase low energy light bulbs) by 2030? | 1 very important 2 fairly important 3 not very important 4 not at all important |
| eb833 | top3_eu_budget_priority2 | And on which of the following would you like EU budget to be spent? Climate change and environmental protection | 0 not mentioned 1 mentioned |
| eb834 | important_renewables4 | How important do you think it is that the (NATIONALITY) government sets targets to increase the amount of renewable energy used, such as wind or solar power, by 2030? | 1 very important 2 fairly important 3 not very important 4 not at all important |
| eb834 | important_reduce4 | How important do you think it is that the (NATIONALITY) government provides support for improving energy efficiency by 2030 (e.g. by encouraging people to insulate their home or buy electric cars)? | 1 very important 2 fairly important 3 not very important 4 not at all important |
| eb871 | support_renewables4 | More public financial support should be given to the transition to clean energies even if it means subsidies to fossil fuels should be reduced | 1 totally agree 2 tend to agree 3 tend to disagree 4 totally disagree |
| eb871 | important_renewables4 | How important do you think it is that the (NATIONALITY) government sets ambitious targets to increase the amount of renewable energy used, such as wind or solar power, by 2030? | 1 very important 2 fairly important 3 not very important 4 not at all important |
| eb871 | important_reduce4 | How important do you think it is that the (NATIONALITY) government provides support for improving energy efficiency by 2030 (e.g. by encouraging people to insulate their home or buy electric cars)? | 1 very important 2 fairly important 3 not very important 4 not at all important |

Table SM3: Climate Change Policy Support Survey Items *(continued)*

| Survey | Survey Item Code | Question Text | Response Categories |
|---|---|---|---|
| eb913 | important_renewables4 | How important do you think it is that the (NATIONALITY) government sets ambitious targets to increase the amount of renewable energy used, such as wind or solar power, by 2030? | 1 very important 2 fairly important 3 not very important 4 not at all important |
| eb913 | important_reduce4 | How important do you think it is that the (NATIONALITY) government provides support for improving energy efficiency by 2030 (e.g. by encouraging people to insulate their home or buy electric cars)? | 1 very important 2 fairly important 3 not very important 4 not at all important |
| eb913 | important_neutral4 | To what extent do you agree or disagree with the following statement: We should reduce greenhouse gas emissions to a minimum while offsetting the remaining emissions, for instance by increasing forested areas, to make the EU economy climate neutral by 2050. | 1 totally agree 2 tend to agree 3 tend to disagree 4 totally disagree |
| eb914 | access_clean4 | The EU must ensure access to clean energy, e.g. encourage a move away from fossil fuels towards energy sources with low greenhouse gas emissions | 1 totally agree 2 tend to agree 3 tend to disagree 4 totally disagree |
| eb951 | support_renewables4 | More public financial support should be given to the transition to clean energies even if it means subsidies to fossil fuels should be reduced | 1 totally agree 2 tend to agree 3 tend to disagree 4 totally disagree |
| eb951 | important_renewables4 | How important do you think it is that the (NATIONALITY) government sets targets to increase the amount of renewable energy used, such as wind or solar power, by 2030? | 1 very important 2 fairly important 3 not very important 4 not at all important |
| eb951 | important_renewables4a | How important do you think it is that the European Union sets targets to increase the amount of renewable energy used, such as wind or solar power, by 2030? | 1 very important 2 fairly important 3 not very important 4 not at all important |
| eb951 | important_reduce4 | How important do you think it is that the (NATIONALITY) government provides support for improving energy efficiency by 2030 (e.g. by encouraging people to insulate their home or buy electric cars)? | 1 very important 2 fairly important 3 not very important 4 not at all important |
| eb951 | important_reduce4a | How important do you think it is that the European Union provides support for improving energy efficiency by 2030 (e.g. by encouraging people to insulate their home or buy electric cars)? | 1 very important 2 fairly important 3 not very important 4 not at all important |
| eb951 | important_neutral4 | We should reduce greenhouse gas emissions to a minimum while offsetting the remaining emissions, for instance by increasing forested areas, to make the EU economy climate-neutral by 2050. | 1 totally agree 2 tend to agree 3 tend to disagree 4 totally disagree |
| eb961 | important_neutral4 | Please tell how important or not important each of the following is: Making Europe the world's first climate-neutral continent by 2050 | 1 very important 2 fairly important 3 not very important 4 not at all important |
| eb961 | important_renewables4b | Please tell how important or not important each of the following is: Increase the share of renewable energy in the European economy and have greater energy efficiency to reduce greenhouse gas emissions. | 1 very important 2 fairly important 3 not very important 4 not at all important |
| eb961 | important_zev4 | Please tell how important or not important each of the following is: Promoting the growth of the zero- and low- emissions vehicles market in order to cut transport emissions. | 1 very important 2 fairly important 3 not very important 4 not at all important |
| eb961 | important_restore4 | Please tell how important or not important each of the following is: Restoring Europe's forests, soils, wetlands and peatlands to increase the absorption of CO2 pollution, making our environment more resilient to climate change | 1 very important 2 fairly important 3 not very important 4 not at all important |

Table SM3: Climate Change Policy Support Survey Items *(continued)*

| Survey | Survey Item Code | Question Text | Response Categories |
|---|---|---|---|
| eb961 | important_renovate4 | Please tell how important or not important each of the following is: Renovating buildings to use more renewable energy and to be more energy efficient | 1 very important 2 fairly important 3 not very important 4 not at all important |
| eb973 | important_neutral4 | Please tell how important or not important each of the following is: Making Europe the world's first climate-neutral continent by 2050 | 1 very important 2 fairly important 3 not very important 4 not at all important |
| eib2020 | regulation2 | The COVID-19 crisis has showed that government measures to combat the pandemic, such as mandatory masks, lockdowns, travel restrictions, etc., together with vaccination, have been successful in containing the spread of the virus. Would you be in favour of stricter government measures imposing changes in people's behaviour to tackle climate change? | 1 no 2 yes |
| eib2021 | regulation2 | The COVID-19 crisis has showed that government measures to combat the pandemic, such as mandatory masks, lockdowns, travel restrictions, etc., together with vaccination, have been successful in containing the spread of the virus. Would you be in favour of stricter government measures imposing changes in people's behaviour to tackle climate change? | 1 no 2 yes |
| eib2022 | regulation2 | Would you be in favour of stricter government measures imposing changes in people's behaviour to tackle climate change? | 1 no 2 yes |
| eib2023 | favor_progressive_carbon_ta | To promote social equity while addressing the climate emergency, some suggest implementing the following new taxation methods. Would you agree with these? A progressive tax on carbon emissions: This means that people who earn more would pay a higher tax rate when they buy things like petrol, to help fight climate change. People who earn less would pay a lower tax rate. The goal is to make sure that everyone pays their fair share to address the climate crisis. | 1 strongly agree 2 agree 3 disagree 4 strongly disagree |
| eib2023 | favor_producer_carbon_tax4 | To promote social equity while addressing the climate emergency, some suggest implementing the following new taxation methods. Would you agree with these? A carbon wealth tax: This would mean that the more people or companies earn from activities that harm the environment, like extracting of using fossil fuels, the higher their taxes on those earnings. The goal is to create incentives to reduce harmful activities and fund efforts to address the climate emergency. | 1 strongly agree 2 agree 3 disagree 4 strongly disagree |
| eib2023 | favor_end_ff_subsidies | To promote social equity while addressing the climate emergency, some suggest implementing the following new taxation methods. Would you agree with these? Fossil fuel tax reform: This involves eliminating subsidies and tax breaks for companies that use fossil fuels and air transport. This additional public money would be used to subsidise renewable energy and clean technologies to fight climate change. The goal is to ensure that public funds are used fairly to address the climate crisis. | 1 strongly agree 2 agree 3 disagree 4 strongly disagree |

Table SM3: Climate Change Policy Support Survey Items *(continued)*

| Survey | Survey Item Code | Question Text | Response Categories |
|---|---|---|---|
| eib2023_latam | regulation2 | Would you be in favour of stricter government measures imposing changes in people's behaviour to tackle climate change? | 1 no 2 yes |
| ess8 | use_coal5 | The highlighted box at the top of this card shows a number of energy sources that can be used to generate electricity. Please take a moment to look over them. How much of the electricity used in [country] should be generated from each energy source? First, how much of the electricity used in [country] should be generated from coal? | 1 a very large amount 2 a large amount 3 a medium amount 4 a small amount 5 none at all |
| ess8 | use_gas5 | How much electricity in [country] should be generated from natural gas | 1 a very large amount 2 a large amount 3 a medium amount 4 a small amount 5 none at all |
| ess8 | use_hydro5 | How much electricity in [country] should be generated from hydroelectric power | 1 a very large amount 2 a large amount 3 a medium amount 4 a small amount 5 none at all |
| ess8 | use_nuclear5 | How much electricity in [country] should be generated from nuclear power | 1 a very large amount 2 a large amount 3 a medium amount 4 a small amount 5 none at all |
| ess8 | use_solar5 | How much electricity in [country] should be generated from solar power | 1 a very large amount 2 a large amount 3 a medium amount 4 a small amount 5 none at all |
| ess8 | use_wind5 | How much electricity in [country] should be generated from wind power | 1 a very large amount 2 a large amount 3 a medium amount 4 a small amount 5 none at all |
| ess8 | use_biomass5 | How much electricity in [country] should be generated from biomass energy | 1 a very large amount 2 a large amount 3 a medium amount 4 a small amount 5 none at all |
| ess8 | fossil_tax5 | To what extent are you in favour or against the following policies in [country] to reduce climate change? Increasing taxes on fossil fuels, such as oil, gas and coal. | 1 strongly in favour 2 somewhat in favour 3 neither in favour nor against 4 somewhat against 5 strongly against |
| ess8 | renewables_subsidy5 | To what extent are you in favour or against the following policies in [country] to reduce climate change? Using public money to subsidise renewable energy such as wind and solar power. | 1 strongly in favour 2 somewhat in favour 3 neither in favour nor against 4 somewhat against 5 strongly against |
| ess8 | appliance_ban5 | To what extent are you in favour or against the following policies in [country] to reduce climate change? A law banning the sale of the least energy efficient household appliances. | 1 strongly in favour 2 somewhat in favour 3 neither in favour nor against 4 somewhat against 5 strongly against |
| lb2010 | cooperate3 | One more question about climate change. Which one of these three statements [HAND CARD] is closest to your opinion: | 1 all countries–including ours–need to work together on climate change 2 those that do the most to cause climate change should find a solution 3 climate change is not that big a problem |
| lits_wave2 | tax2 | Would you be willing to give part of your income or pay more taxes, if you were sure that the extra money was used to…combat climate change | 1 yes 2 no |
| pew2015 | reduce_emissions2 | Countries from around the world will meet in December in Paris to deal with global climate change. They will discuss an agreement to limit greenhouse gas emissions, such as from burning coal or [gas/petrol]. Do you support or oppose (survey country) limiting its greenhouse gas emissions as part of such an agreement? | 1 support 2 oppose |

Table SM3: Climate Change Policy Support Survey Items *(continued)*

| Survey | Survey Item Code | Question Text | Response Categories |
|---|---|---|---|
| pipa2009 | reduce_emissions2 | As you may know, [country] and other countries from around the world will be meeting in December in Copenhagen to develop a new agreement to take steps against climate change by limiting greenhouse gas emissions. If the other countries come to an agreement, do you think [country] should or should not be willing to commit to limiting its greenhouse gas emissions as part of such an agreement | 1 should be willing 2 should not be willing |
| pipa2009 | expand_forest4 | Would you favor or oppose [country] taking each of the following steps to help deal with climate change? :Preserving or expanding forested areas, even if this means less land for agriculture or construction | 1 favor strongly 2 favor somewhat 3 oppose somewhat 4 oppose strongly |
| pipa2009 | reduce_coal4 | Limiting the rate of constructing coal-fired power plants, even if this increases the cost of energy | 1 favor strongly 2 favor somewhat 3 oppose somewhat 4 oppose strongly |
| pipa2009 | increase_milage4 | Gradually increasing the requirements for fuel efficiency in automobiles, even if this raises the cost of cars and bus fares | 1 favor strongly 2 favor somewhat 3 oppose somewhat 4 oppose strongly |
| pipa2009 | reduce_subsidies4 | Gradually reducing government subsidies that favor private transportation, even if this raises its cost | 1 favor strongly 2 favor somewhat 3 oppose somewhat 4 oppose strongly |
| pipa201007 | reduce_emissions2 | As you may know, [country] and other countries from around the world will be meeting in December in Copenhagen to develop a new agreement to take steps against climate change by limiting greenhouse gas emissions. If the other countries come to an agreement, do you think [country] should or should not be willing to commit to limiting its greenhouse gas emissions as part of such an agreement | 1 should be willing 2 should not be willing |
| pipa201007 | expand_forest4 | Would you favor or oppose [country] taking each of the following steps to help deal with climate change? :Preserving or expanding forested areas, even if this means less land for agriculture or construction | 1 favor strongly 2 favor somewhat 3 oppose somewhat 4 oppose strongly |
| pipa201007 | reduce_coal4 | Limiting the rate of constructing coal-fired power plants, even if this increases the cost of energy | 1 favor strongly 2 favor somewhat 3 oppose somewhat 4 oppose strongly |
| pipa201007 | increase_milage4 | Gradually increasing the requirements for fuel efficiency in automobiles, even if this raises the cost of cars and bus fares | 1 favor strongly 2 favor somewhat 3 oppose somewhat 4 oppose strongly |
| pipa201007 | reduce_subsidies4 | Gradually reducing government subsidies that favor private transportation, even if this raises its cost | 1 favor strongly 2 favor somewhat 3 oppose somewhat 4 oppose strongly |

* Survey dataset codes correspond to those used in the DCPOtools R package (Solt, Hu, and Tai 2019).

# B Meta-Regression of Climate Concern

The studies consolidated by the large-scale SPMs presented in the main text parallel those presented in Bush and Clayton (2023): each is a two-level model of respondents nested in countries for one item drawn from a single cross-national survey predicted by gender, economic development, and their interaction. This approach has the advantage of making clear that any difference in conclusions is the result of the broader data brought to bear on the question rather than any change in the research strategy, but it does come at some sacrifice to statistical power relative to a strategy that treats each available country-year-survey-item as a study of the relationship between gender and climate attitudes and then uses meta-regression (see, e.g., Higgins, López-López, and Aloe 2020) to test whether the differences across these studies can be explained with differences in economic development. In light of the null and weak findings on the interaction between gender and economic development in predicting, respectively, perceptions of benefits versus costs of climate mitigation efforts and support for policies to fight climate change, the main text took advantage of the increased power of this latter approach to better discern any relationships among these variables. This appendix presents a meta-regression for climate concern that follows this same procedure.

Each of these studies examined here, then, consists of the correlation between the indicator variable for women and responses to one item tapping climate change concern from one survey asked in one country in one year with the Fisher's $z$ transformation applied. These correlations are then predicted using logged GDP/capita in multilevel models that include varying intercepts to capture the lack of independence between observations from the same survey, those employing the same item, and those of the same country.

Figure SM2 presents the results. These studies are plotted in the left panel: the correlation between respondents' gender and their climate concern expressed in a single survey-item in a single country-year appears on the $y$-axis; the vertical whiskers on each point represent uncertainty in the correlation due to sampling error. The economic development of the country at the time of the survey, measured as in Bush and Clayton (2023) as the log of GDP/capita, is on the $x$-axis. The right panel of Figure SM2 presents the posterior probability distribution from a meta-regression of these correlations on economic development. This distribution provide additional support for the first claim of Bush and Clayton (2023) that the gender gap in climate concern increases with economic development like that reported in Figure 1 in the main text, with a very similar median value and even tighter credible intervals and bounds that entirely exclude zero. Women express more concern about climate change than men where GDP/capita is higher.

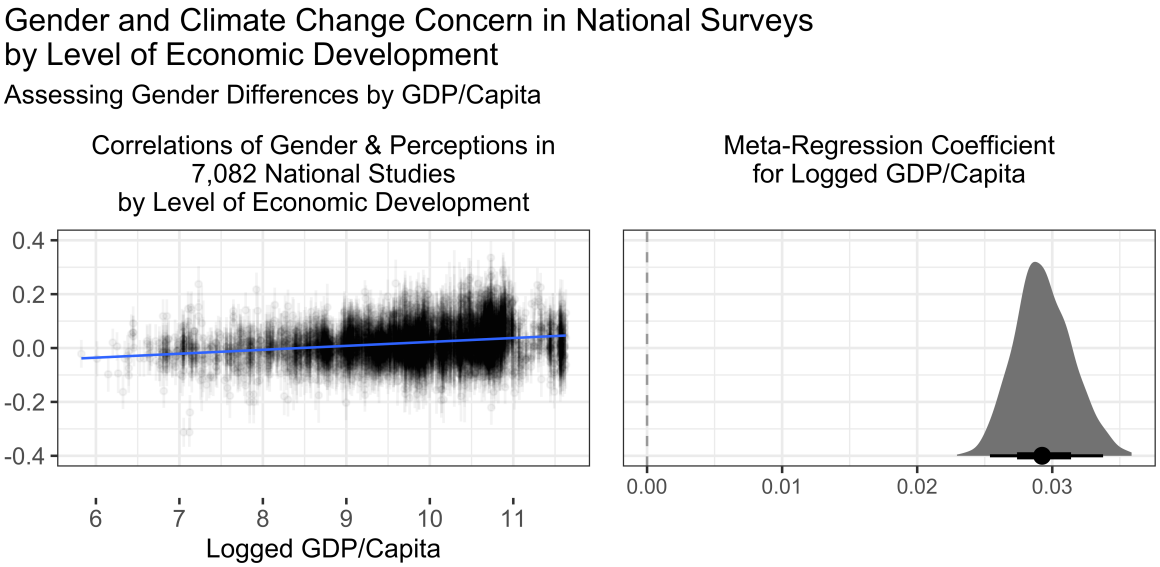



Figure SM2: The Difference Between Women and Men in Climate Concern Is Generally Larger in Richer Countries Than in Poorer Countries

# References

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